


## Title

*Say (verbatim):*

- Good morning, everyone. Thank you for being here. My name is Jorge Mayorga, from Universidad de los Andes.
- I want to open with a question that sounds simple but really is not: in a power grid that is increasingly run by inverters, which frequency estimator should we actually trust?
- We tend to assume the estimator is a solved, interchangeable block. It is not.
- The honest answer to that question is, it depends, and the whole point of this talk is to make 'it depends' precise, measurable, and useful for engineers.

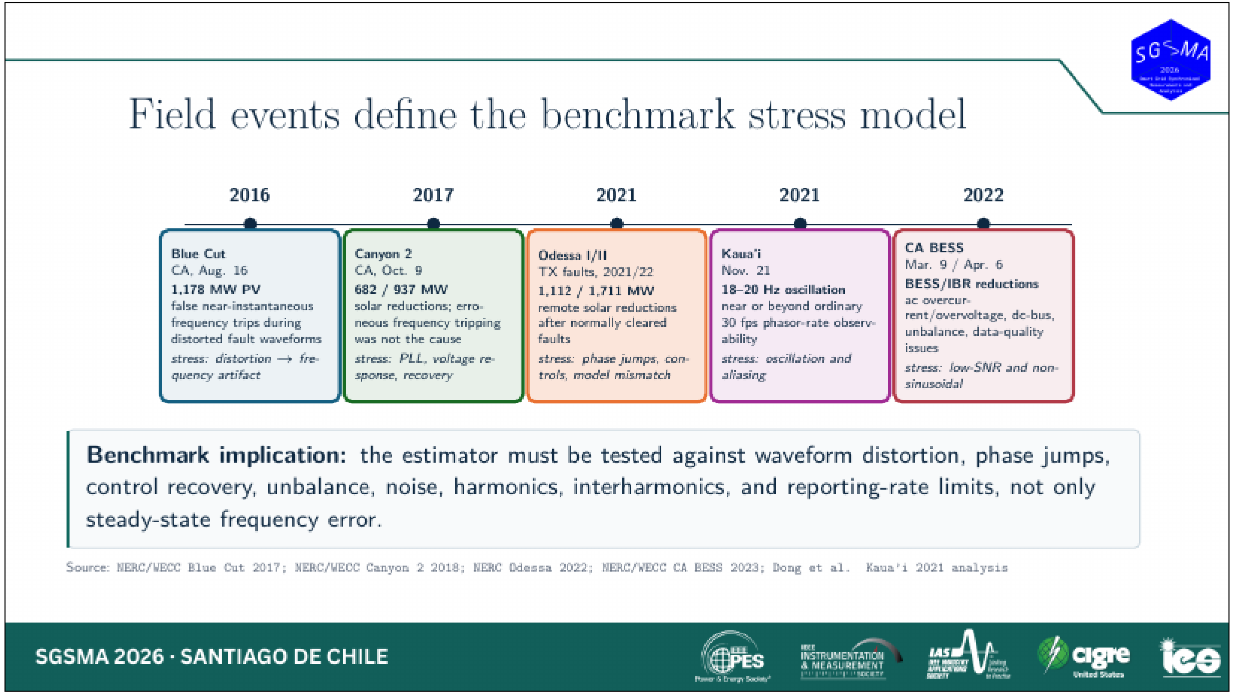
→ *So let me begin with why this is a problem worth solving.*

**Key line:** There is no universal best estimator; the right choice depends on what you measure.




# Motivation (section)

**Key line:** Why this matters.






# Field events define the stress model

*Say (verbatim):*

- Let me ground this in reality. Real inverter-driven disturbances look nothing like the clean sine wave we use in textbooks.
- If you look at recorded events, Blue Cut, Odessa, the Kauai oscillation, you see phase jumps, fast rate-of-change of frequency, harmonics, and noise all arriving at the same time.
- So instead of inventing convenient test signals, we let these real field events define the stress cases that our benchmark has to cover.
- That keeps the evaluation honest, because the grid does not care about our assumptions.

→ *And here is the catch: the very instrument we use to watch these events also shapes what we conclude about them.*

**Key line:** Field events, not textbook signals, set the bar.

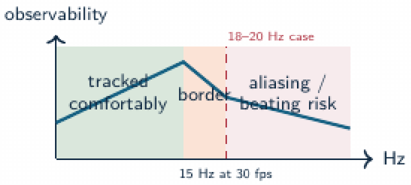
# Standard PMU streams show consequences, not always causes

### Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

### Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.




### Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.





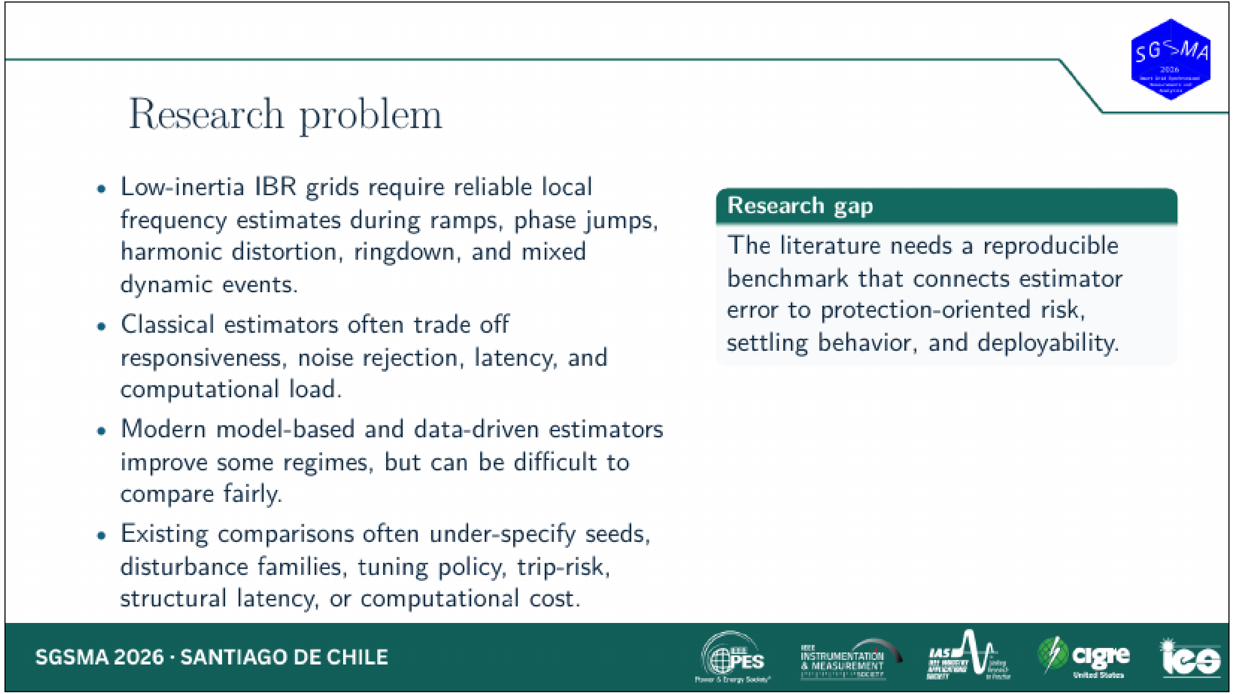
## PMU streams show consequences

*Say (verbatim):*

- A standard PMU hands us a frequency value and a rate-of-change of frequency.
- But during a fast inverter event, those numbers show the consequence, not always the cause.
- And crucially, it is the estimator buried inside the PMU that decides what we actually see on the screen.
- Two different estimators, looking at the very same event, can report two different stories.

→ *So before we can trust any analysis of an event, we first have to ask which estimator produced those numbers.*

**Key line:** Choosing the estimator is itself a measurement decision.





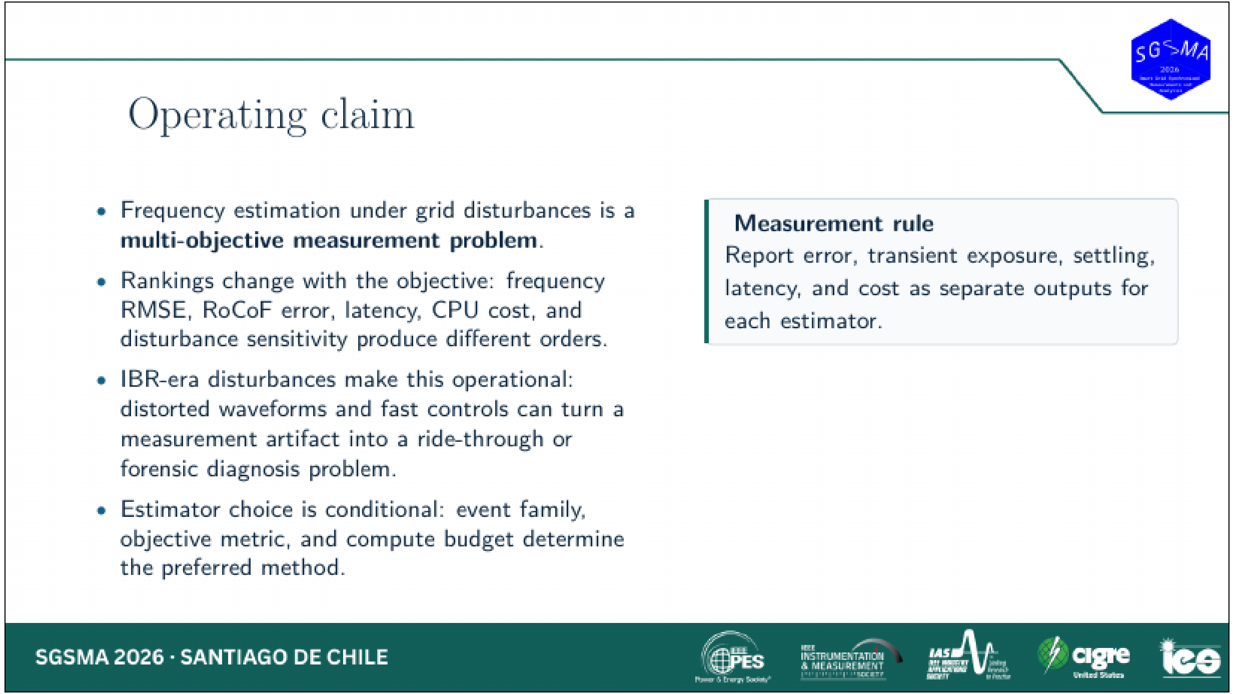

## Research problem

*Say (verbatim):*

- And that points straight at the gap we are addressing. Today there is no agreed, systematic way to compare these estimators under inverter-specific stress.
- The compliance tests we do have run on clean, well-behaved signals.
- So they tell us a method passes a standard, but they never tell us how it behaves during a real, messy, composite event.
- We are essentially certifying estimators on the easy cases and hoping they survive the hard ones.

→ *Our response to that gap is one clear claim.*

**Key line:** We need a fair, inverter-relevant way to compare estimators.





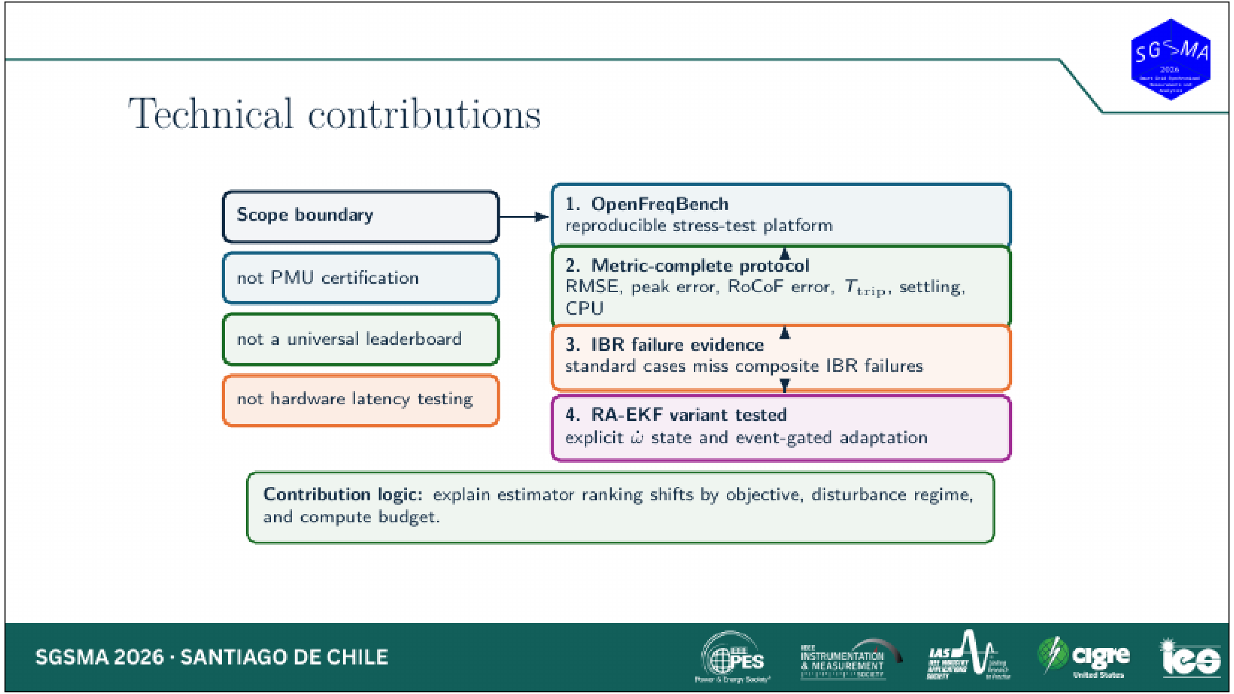

# Operating claim

*Say (verbatim):*

- The central claim of this work is that estimator selection has to be conditional.
- It is not a single ranking. It depends on three things at once: the type of disturbance, the metric you actually care about, and your compute budget.
- Change any one of those, and the best estimator can change with it.
- So our job is not to crown a winner; it is to map out which method wins under which conditions.

→ *And to back that claim with evidence, we built three things.*

**Key line:** There is no single best estimator, only a best one for a given job.





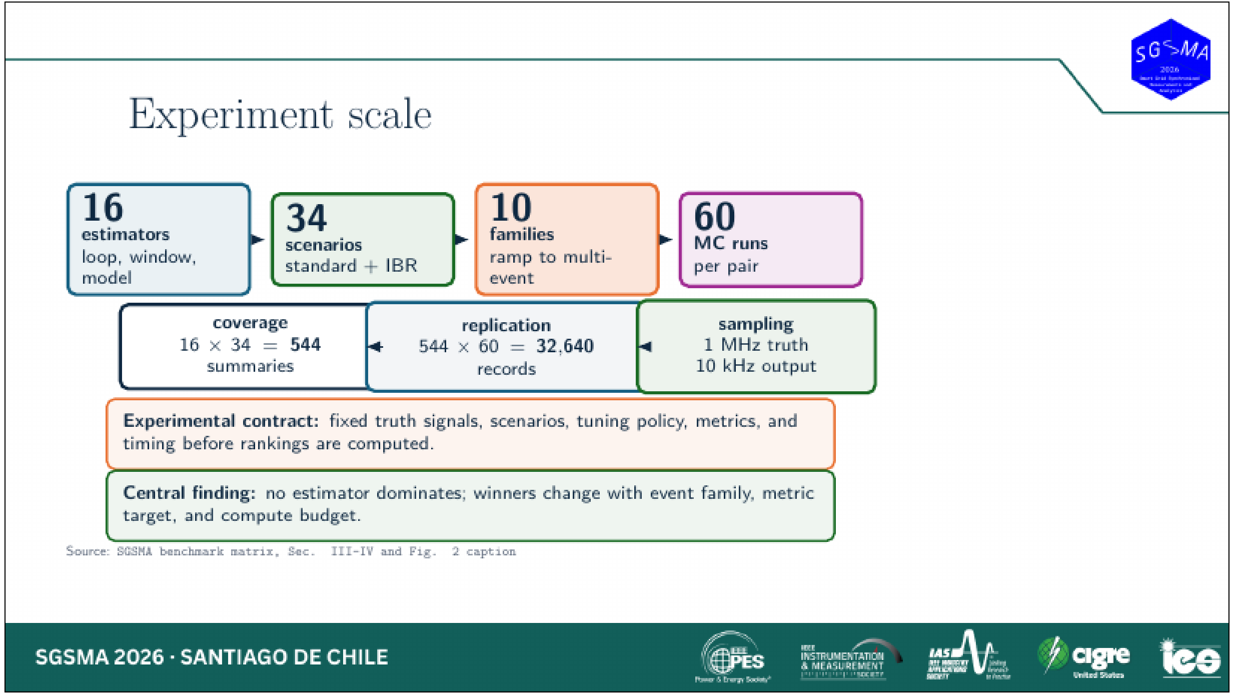

## Technical contributions

*Say (verbatim):*

- We make three contributions. First, an open benchmark platform built around known-truth scenarios, so the error is never in doubt.
- Second, a common execution contract, so that all eighteen estimators are tested under exactly identical conditions, with no method getting an unfair advantage.
- And third, a metric vector that keeps accuracy, trip-risk, latency, and computational cost as separate, explicit outputs, instead of collapsing them into one number.

→ *Let me give you a feel for the scale of all this.*

**Key line:** Open platform, common contract, and multi-metric evidence.






## Experiment scale

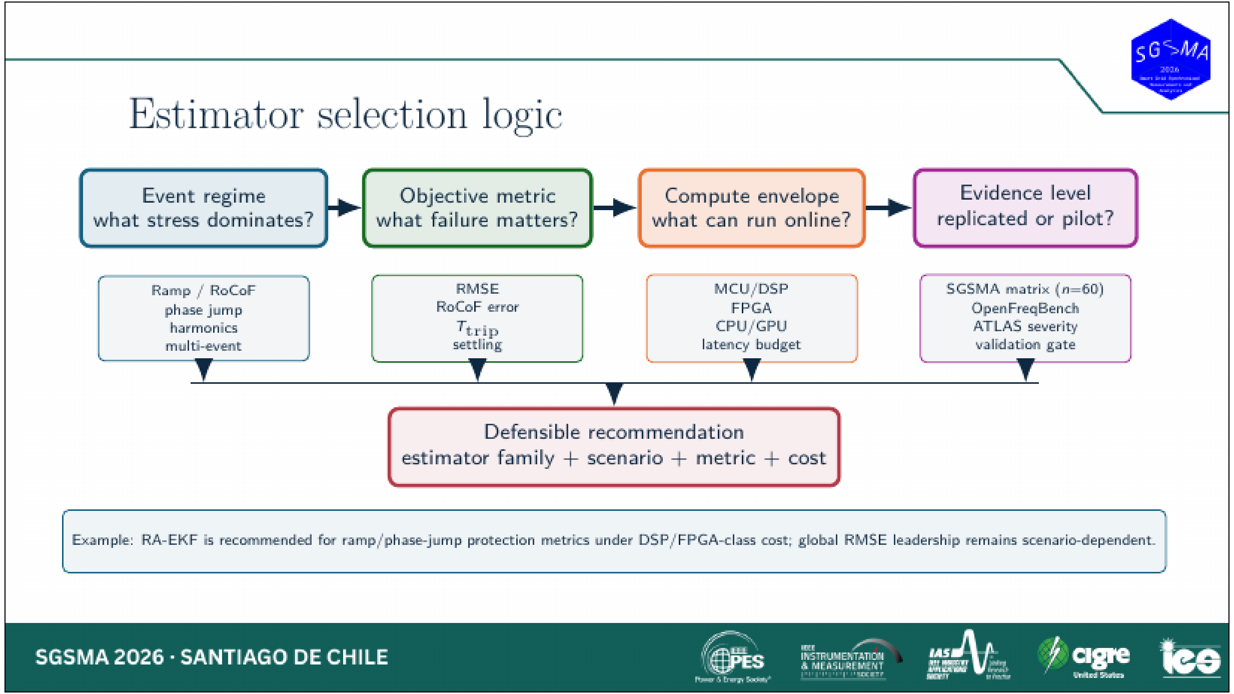
*Say (verbatim):*

- The scale is deliberately large: eighteen estimators, thirty-three scenarios, across five families of stress, all with Monte-Carlo repetitions.
- On top of that, we built a severity-sweep engine that we call ATLAS, which pushes each stress from mild to extreme.
- And every single result is reproducible from saved artifacts, so anyone can regenerate the figures you will see today.

→ *And the logic for choosing among all of these methods actually fits into a single picture.*

**Key line:** Large, factorial, and fully reproducible.

## Estimator selection logic (build-up)




*Say (verbatim):*

- This diagram is, in a sense, the entire talk in miniature.
- Selection comes down to three questions: which event class are you facing, which metric matters for your application, and what compute budget do you have.
- Watch as each question appears, because each one, on its own, can send you to a different estimator.

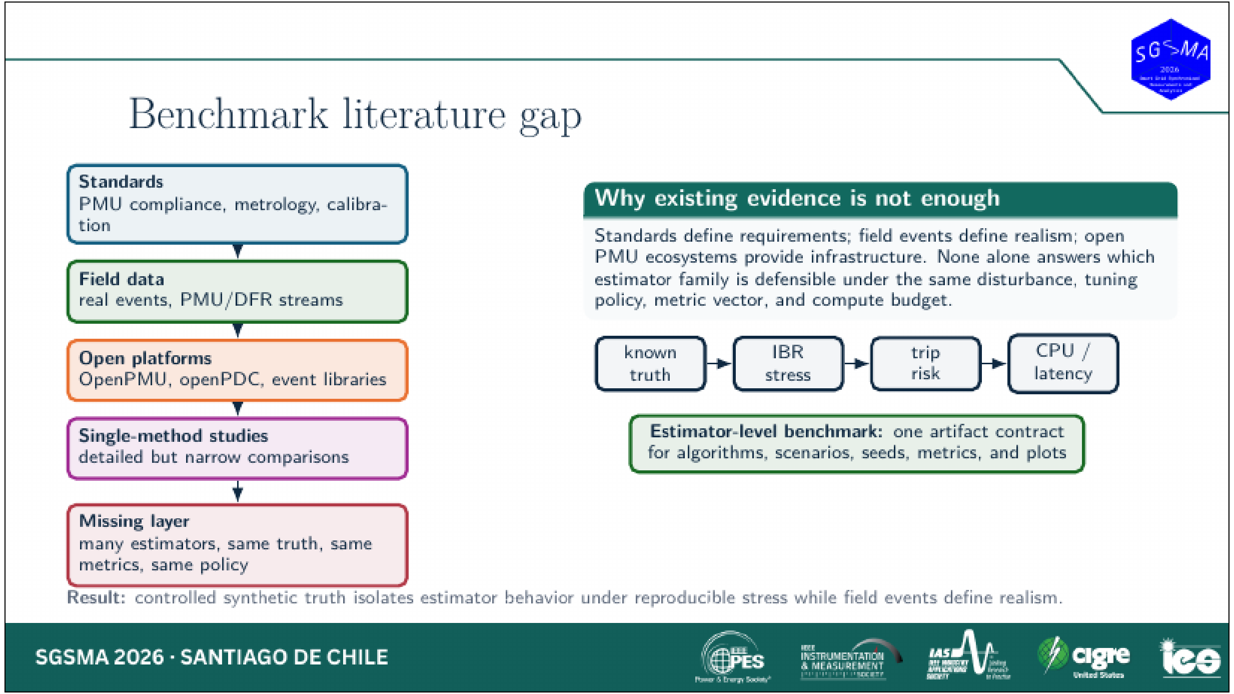
→ *With that logic in hand, let me now show you the platform itself.*

**Key line:** This one diagram is the whole talk in miniature.




# Benchmark and platform (section)

**Key line:** First, the benchmark and the platform.






# Benchmark literature gap

*Say (verbatim):*

- If you look at the existing literature, most studies compare a handful of estimators on just one or two cases.
- What has been missing is a benchmark that covers the full estimator space, against composite inverter stress, with metrics you can actually reproduce.
- That combination, breadth, realism, and reproducibility, simply did not exist.

→ *So that is exactly what we set out to build.*

**Key line:** That blank space on the map is the gap we fill.

# Benchmark comparison axes

| Evidence source | Known truth | IBR composite stress | Trip-risk | CPU/latency | Reproducible artifacts |
|---|---|---|---|---|---|
| IEC/IEEE-style compliance | Yes | Limited | No | No | Device-dependent |
| Field PMU/DFR data | No | Yes | Possible | No | Dataset-dependent |
| Single-method studies | Often | Partial | Partial | Partial | Often limited |
| Estimator benchmark | Yes | Yes | Yes | Yes | Yes |

**Measured outputs**

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

**Interpretive effect**

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.





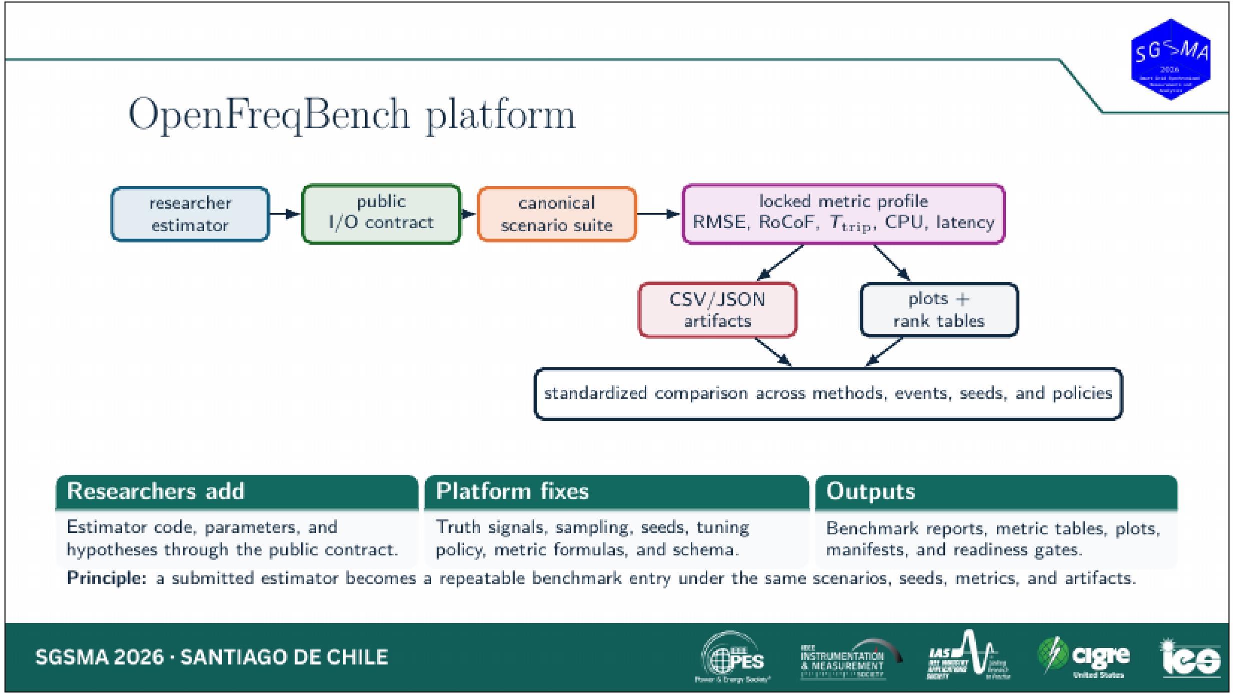
## Benchmark comparison axes

*Say (verbatim):*

- We compare the methods on four axes at the same time: accuracy, robustness, computational cost, and protection trip exposure.
- And the whole reason we keep them separate is that a method can win brilliantly on one axis and quietly lose on another.
- A single leaderboard would hide exactly the trade-off that an engineer needs to see.

→ *Here is the platform that measures all four of them.*

**Key line:** One number hides the trade-off; four axes reveal it.






# OpenFreqBench platform

*Say (verbatim):*

- This is OpenFreqBench. It has four parts.
- A scenario factory that generates waveforms with known true frequency. A uniform interface so every estimator plugs in the same way. A locked metric engine so the scoring never changes between methods. And fully traceable artifacts, every CSV and every figure.
- The design philosophy is simple: known truth in, locked metrics out, and everything traceable.

→ *Let me show you concretely how we keep the comparison fair.*

**Key line:** Known truth in, locked metrics out, everything traced.



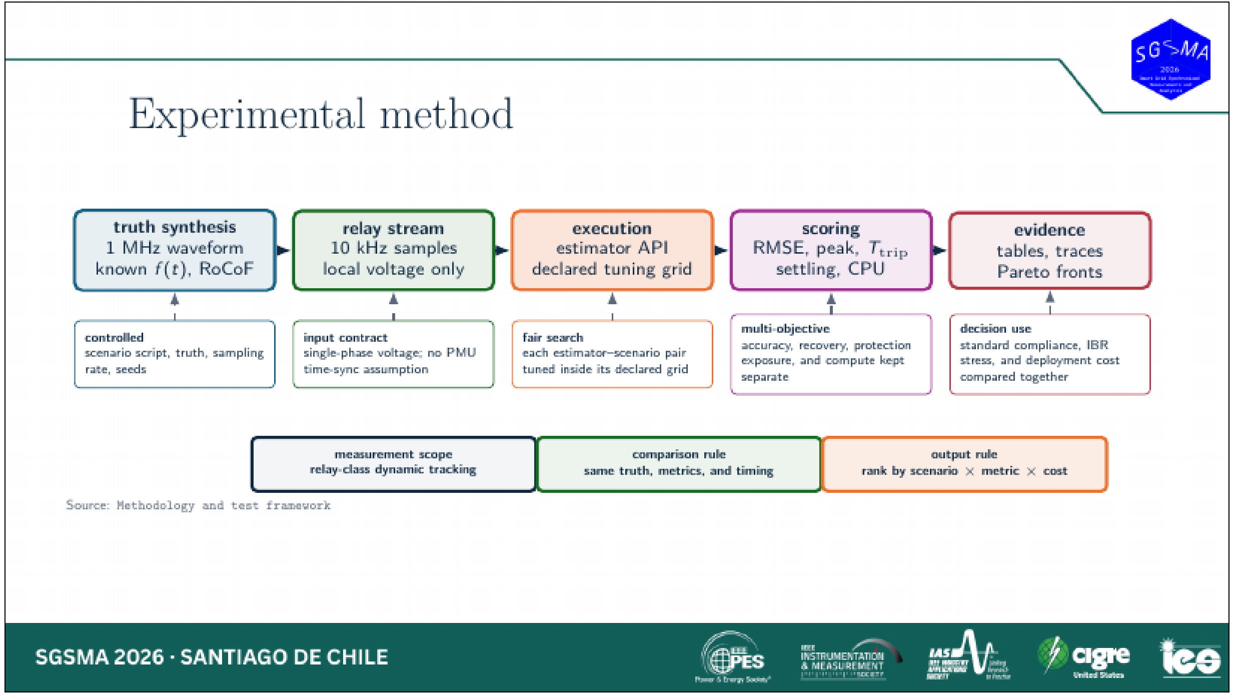

# Experimental method (1)

*Say (verbatim):*

- Fairness is really the heart of this whole project.
- Every estimator sees the exact same input trace, driven by the same random seeds, with the same warm-up handling.
- Nobody gets a cleaner signal or a luckier start. The comparison is fair by construction, not because we trust ourselves to be fair.

→ *And we are just as strict about tuning and about timing.*

**Key line:** Fairness is enforced by construction, not by trust.





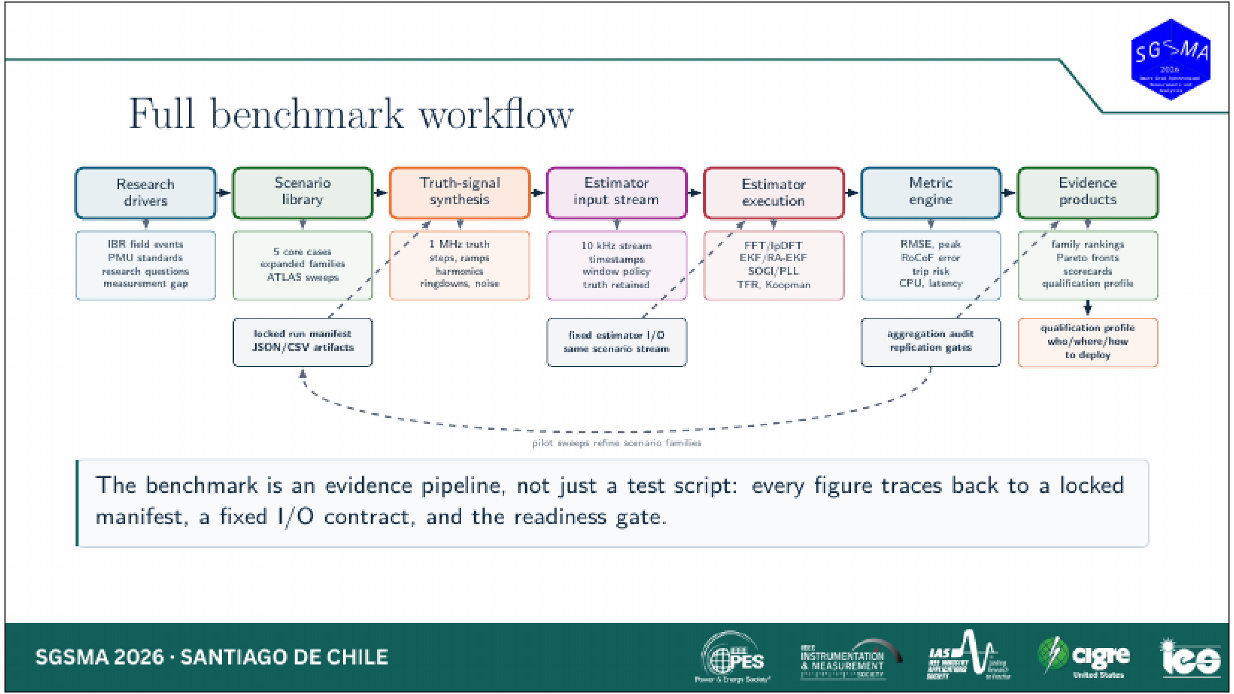

## Experimental method (2)

*Say (verbatim):*

- We tune every method under a single fixed policy, we discard the warm-up region before scoring, and we record CPU cost as a genuine output, not as an afterthought.
- So the computational numbers you will see are measured, not assumed.

→ *Put all of that together, and this is the pipeline.*

**Key line:** Nothing is hand-tuned to flatter one method.





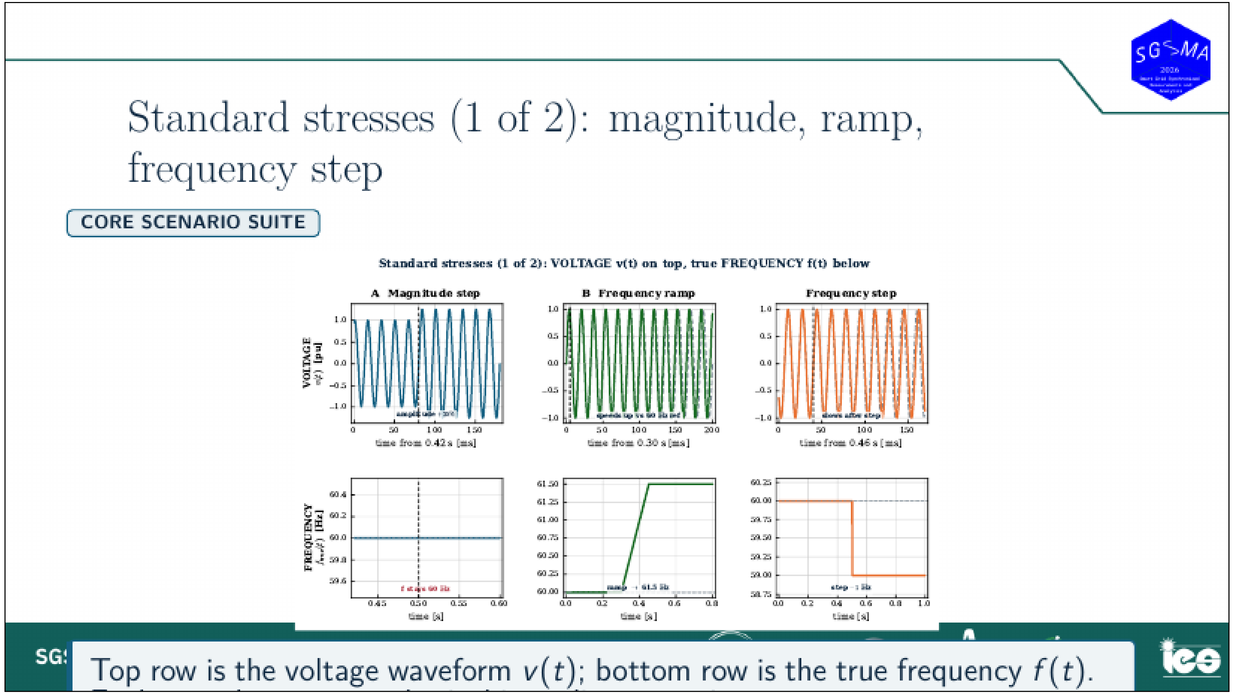

# Full benchmark workflow

*Say (verbatim):*

- The workflow runs in a fixed order: scenario, then Monte-Carlo, then the estimator, then the locked metrics, and finally the artifacts.
- And I want to stress this: every single figure in this talk traces straight back to those artifacts. Nothing here is hand-drawn.

→ *Now let me introduce the scenarios themselves.*

**Key line:** One reproducible pipeline, end to end.





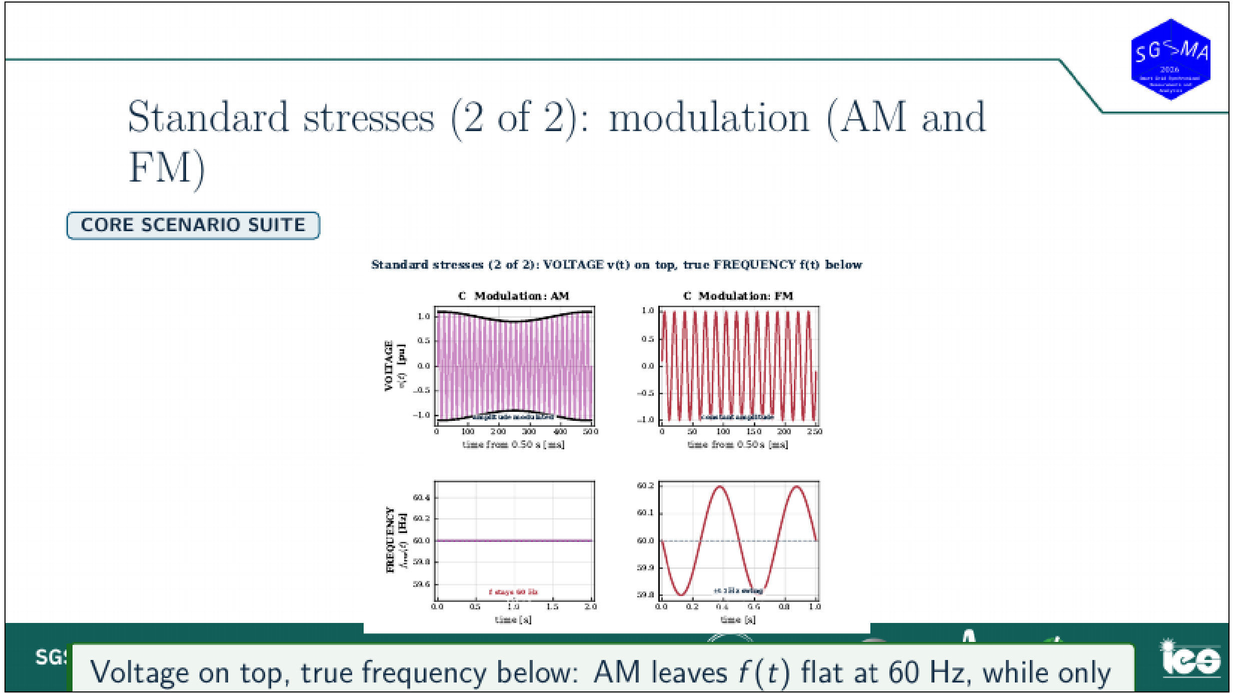

## Standard stresses (1 of 2): magnitude, ramp, step

*Say (verbatim):*

- We start with the standard stresses. On this slide the top row is the voltage waveform, and the bottom row is the true frequency, for three cases: a magnitude step, a frequency ramp, and a frequency step.
- Because we generate the waveform ourselves, we know that true frequency at every single sample, so the error is exact.
- And notice the magnitude step on the left: the voltage clearly changes, but the frequency below it stays flat at sixty hertz.

→ *Modulation gives us the other two standard cases.*

**Key line:** Top row is voltage; bottom row is the true frequency we score against.



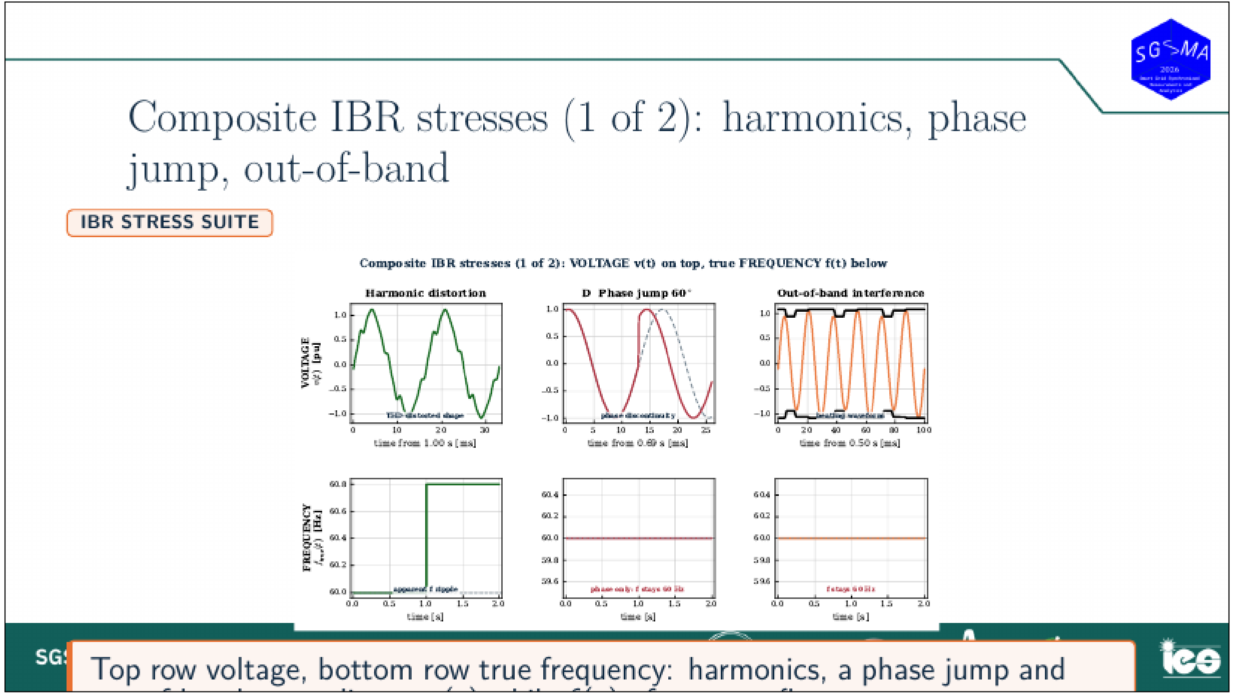

# Standard stresses (2 of 2): modulation AM and FM

*Say (verbatim):*

- Here are the two modulation cases, again voltage on top and true frequency below.
- Amplitude modulation, on the left, leaves the frequency perfectly flat at sixty hertz. Only frequency modulation, on the right, actually moves it.
- So an estimator can be disturbed by a signal whose true frequency never changed at all.

→ *Now let us make the scenarios composite and realistic.*

**Key line:** AM leaves the frequency flat; only FM truly moves it.






# Composite IBR stresses (1 of 2): harmonics, jump, OOB

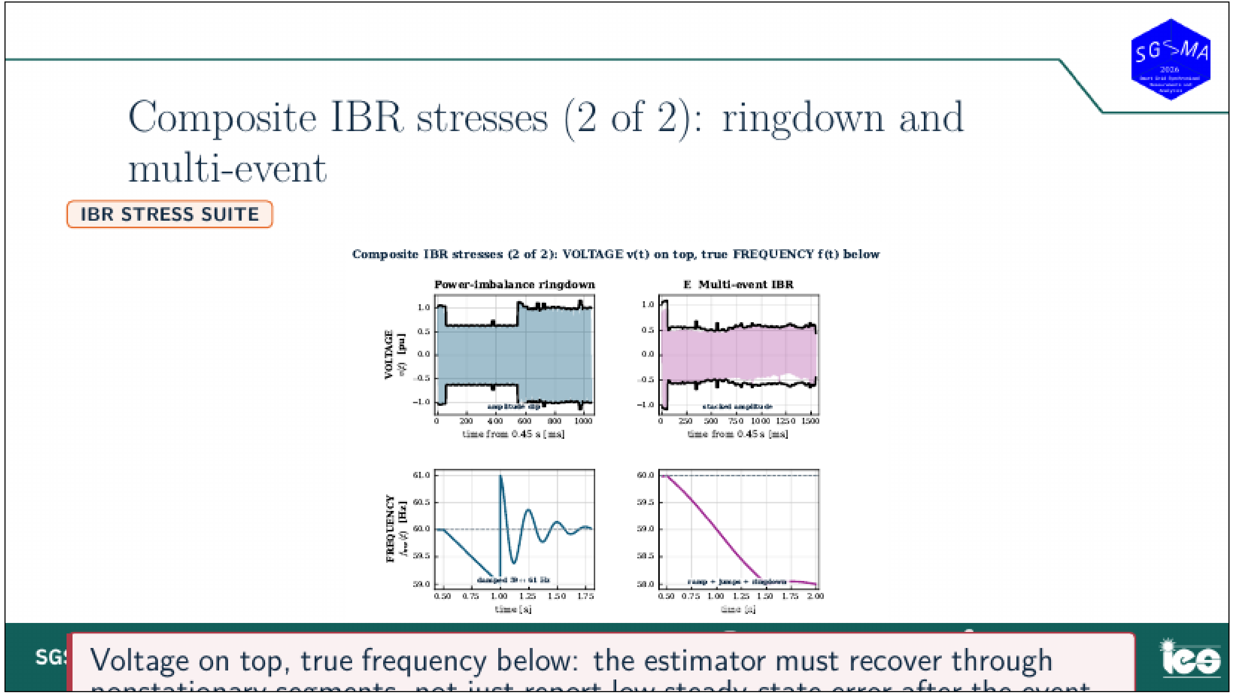
*Say (verbatim):*

- Now we stack the physics. Top row voltage, bottom row true frequency, for three composite cases: harmonic distortion, a phase jump, and out-of-band interference.
- These badly distort the voltage waveform, and yet in several of them the true frequency stays essentially flat.
- That gap, an ugly voltage but a calm true frequency, is exactly what trips up the wrong estimator.

→ *The last two composite cases are the genuinely nonstationary ones.*

**Key line:** A distorted voltage does not mean the true frequency moved.

# Composite IBR stresses (2 of 2): ringdown, multi-event

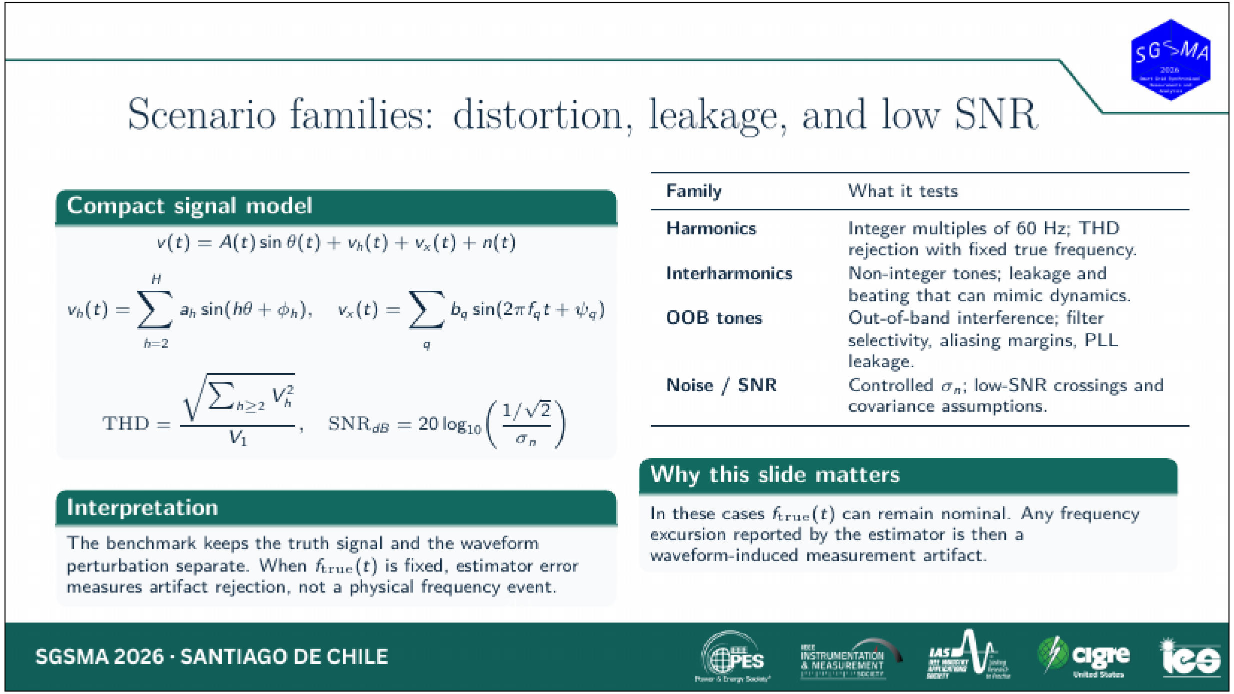



*Say (verbatim):*

- And here are the two hardest cases: a power-imbalance ringdown, and the stacked multi-event sequence, voltage on top and true frequency below.
- Now the frequency really does move, through damped oscillations and then a whole chain of events.
- The estimator has to track through these nonstationary segments, not just settle down afterward.

→ *Before the methods, let me name the three mechanisms that actually break estimators.*

**Key line:** Recovery through nonstationary segments is the real test.






## Scenario families: distortion, leakage, and low SNR

*Say (verbatim):*

- It helps to group the hard scenarios into three families, because each one breaks an estimator in a different way.
- Harmonic distortion attacks any method that assumes a pure sinusoid. Spectral leakage hurts the window-based methods when the frequency does not sit neatly inside a bin.
- And a low signal-to-noise ratio simply drowns the zero-crossing and timing-based methods.
- Keeping these three failure mechanisms separate is exactly what lets us explain, later, why a given method wins or collapses.

→ *With those mechanisms named, here are the estimators we put up against them.*

**Key line:** Distortion, leakage, and low SNR are three distinct failure mechanisms, not one.

# Estimator families

**Loop-based**
PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

**Model-based / stochastic**
EKF, UKF, RA-EKF, LKF, LKF2.

**Legacy baselines**
TKEO and ZCD are retained as comparison references outside the recommended deployment set.

**Window / spectral**
IpDFT, TFT, ESPRIT, Prony, MUSIC.

**Adaptive and data-driven**
RLS, Koopman (RK-DPMU).

**Interpretation rule**
These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.





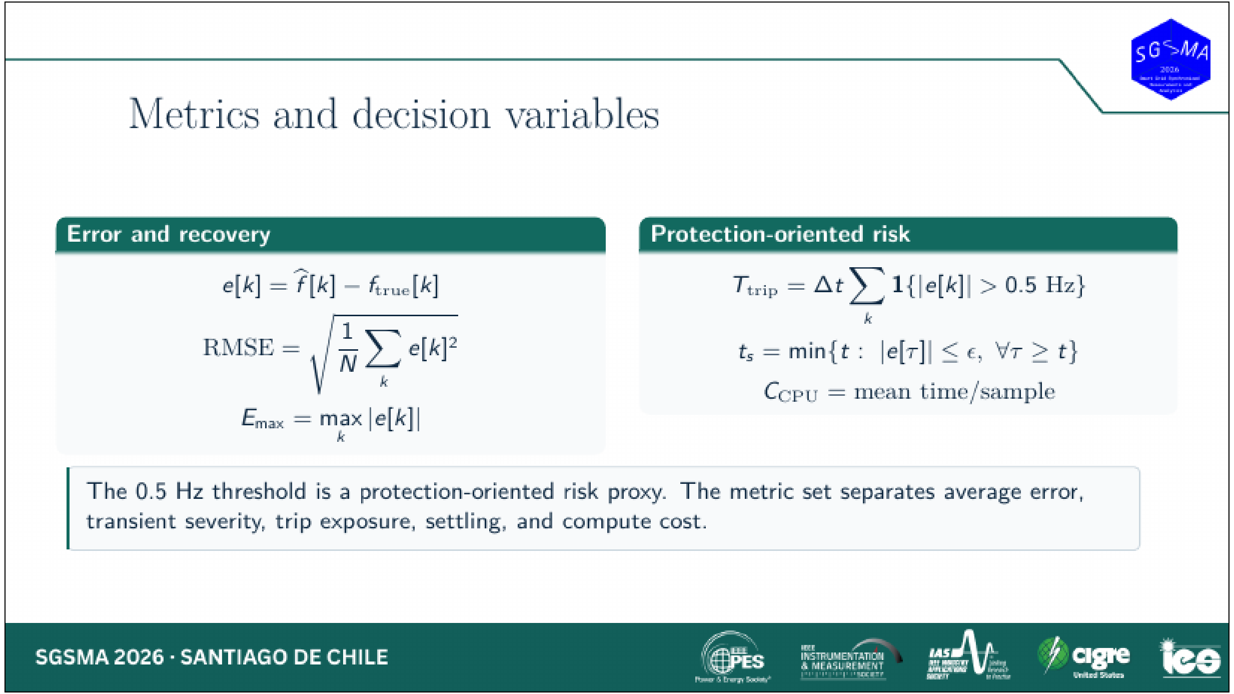
## Estimator families

*Say (verbatim):*

- The eighteen estimators sort naturally into families.
- Loop-based methods, the PLLs. Window and spectral methods. Model-based Kalman filters. Adaptive methods. And data-driven methods.
- We deliberately analyze by family, rather than method by method, because the family is what really explains the behavior we see, the strengths and the failure modes.

→ *And these are the numbers we score them on.*

**Key line:** Eighteen estimators, five families, one fair contract.






# Metrics and decision variables

*Say (verbatim):*

- Here are the metrics. RMSE captures the average error. Peak error captures the worst moment.
- And then a trip-risk time, which counts how long the error stays above half a hertz, because that is where protection relays start to act.
- Plus settling time, and CPU cost. Five dimensions, kept separate on purpose.

→ *Now let me show you what the data actually says.*

**Key line:** Half a hertz is our protection-oriented risk proxy.


# Core evidence (section)

**Key line:** The core evidence: five scenarios, from standard to composite.

## Standard scenario results (build-up)

*Say (verbatim):*

- Let us start with the standard cases. On the step and the modulation, almost everything looks excellent, all of them below fifty millihertz.
- If we stopped here, we would conclude that the estimator choice barely matters.
- CLICK. But now watch the ramp column. The textbook EKF and the IpDFT intermittently blow up, to thirty-seven and fourteen hertz.
- And these are not edge methods; these are textbook, widely-used estimators.

→ *You would expect a phase jump to be even worse, so let me show you that next.*

**Key line:** Robustness, not point accuracy, is what separates the families.

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# Standard scenario results

| Method | Step | Ramp | Mod. |
|---|---|---|---|
| **RA-EKF** | 0.000 | **0.006** | 0.037 |
| **SOGI-FLL** | 0.005 | 0.013 | 0.040 |
| **Koopman** | 0.003 | 0.017 | 0.044 |
| **TFT** | 0.001 | 0.019 | 0.046 |
| **UKF** | 0.000 | 0.017 | 0.055 |
| **LKF2** | 0.138 | 0.015 | **0.028** |
| **PLL** | 0.033 | **1.47** | **0.227** |
| **IpDFT** | 0.000 | **14.4** | 0.114 |
| **EKF** | 0.000 | **37.1** | 0.064 |

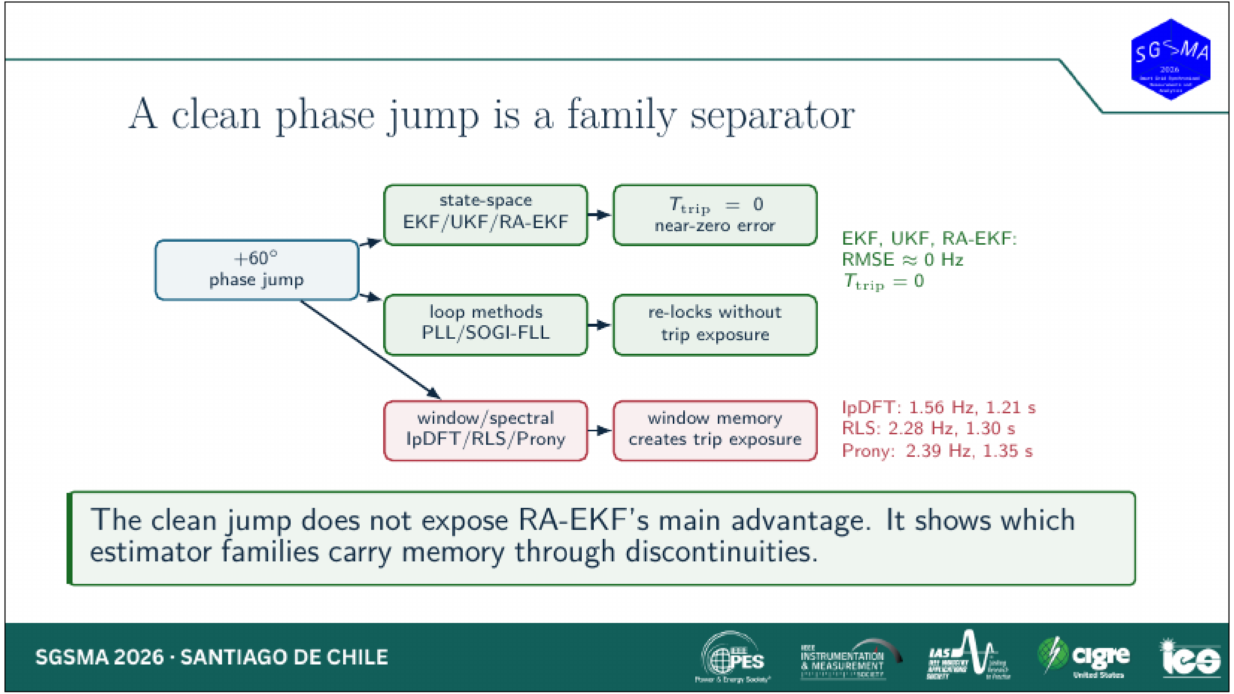
RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

**What the heatmap shows**

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

**Operational consequence**

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.





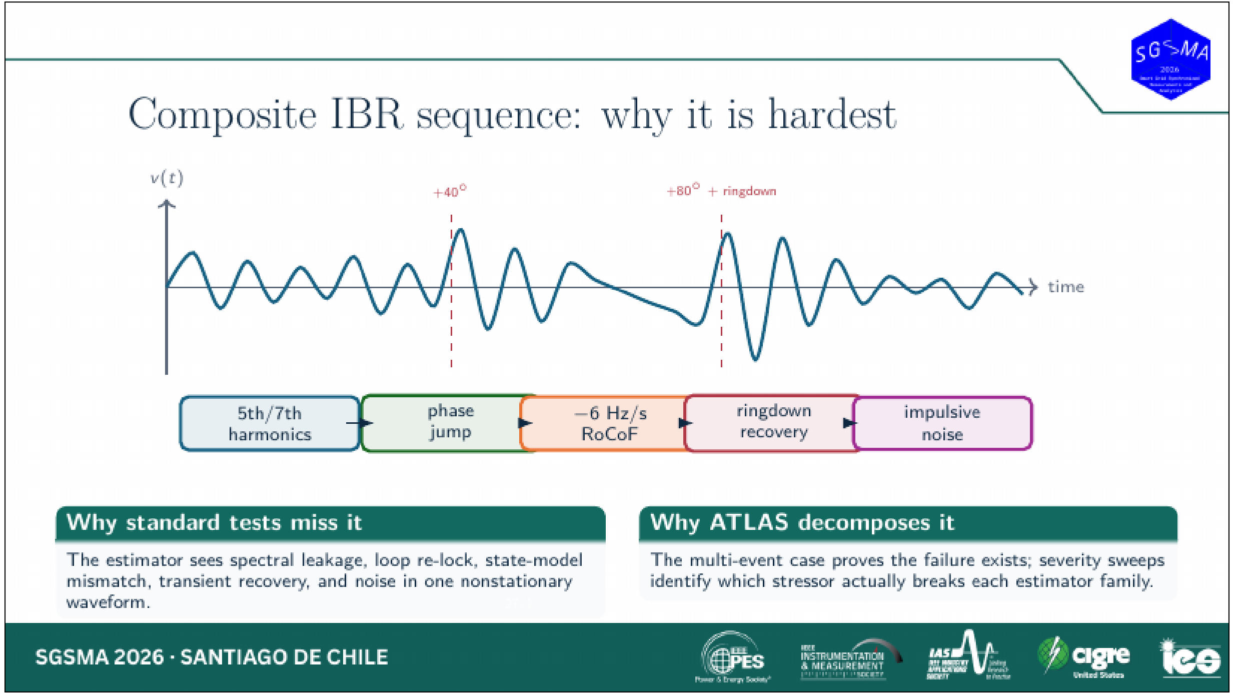

## Scenario D: phase jump separates families

*Say (verbatim):*

- A clean sixty-degree phase jump cleanly separates the families.
- The state-space filters and the PLLs absorb it gracefully, with effectively zero trip exposure.
- The windowed and spectral methods, on the other hand, carry over a full second of trip exposure, because their analysis window straddles the discontinuity.
- But here is the part that surprised us.

→ *Because a single, clean jump is not actually the hard problem.*

**Key line:** The phase jump does not break the EKF; divergence on the ramp is the real danger.






# Composite IBR sequence: hardest case

*Say (verbatim):*

- Scenario E stacks everything together: harmonics, two phase jumps, a rate-of-change segment, a ringdown, and impulsive noise.
- So spectral leakage, loop re-locking, and state-model mismatch all hit the estimator at the very same instant.
- There is nowhere for a method to hide.

→ *And this is exactly where the leaderboard falls apart.*

**Key line:** This is the hardest, most realistic case in the whole suite.

## Scenario E: the leaderboard breaks

| Method | RMSE [Hz] | $T_{trip}$ [s] | CPU [$\mu$s] |
|---|---|---|---|
| UKF | **1.09** | **0.78** | 14 |
| EKF | 1.13 | 0.87 | 11 |
| SOGI-PLL | 1.14 | **0.78** | 11 |
| SOGI-FLL | 1.14 | 0.84 | **9** |
| IpDFT | 1.14 | 1.35 | 17 |
| RA-EKF | 1.16 | 0.92 | 15 |
| Koopman | 1.90 | 0.84 | **15 241** |
| ZCD | **426** | 1.55 | 7 |

**No accurate winner**

Under the stacked sequence every usable method collapses to $\approx$ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

**Scalar ranking failure**

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000$\times$ costlier; RA-EKF sits mid-pack but never diverges.

Source: full_mc_benchmark, $n = 5$: IBR_Multi_Event





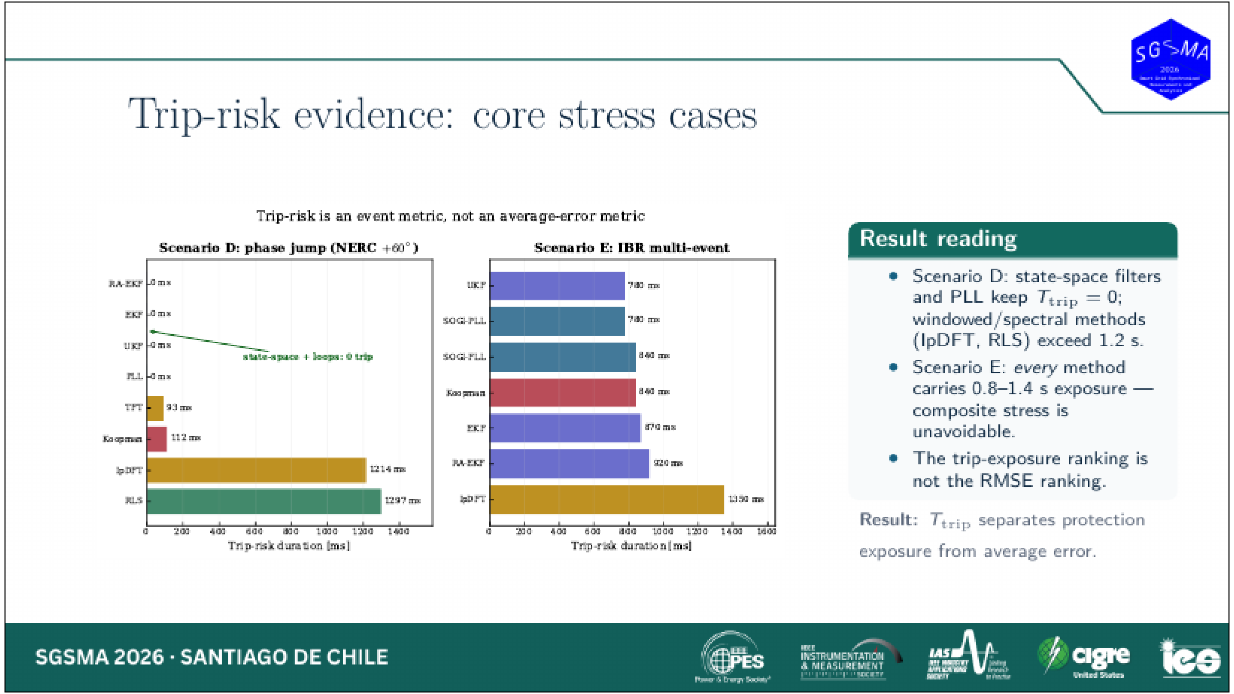
# Scenario E: the leaderboard breaks

*Say (verbatim):*

- Under that stacked sequence, every usable method collapses to roughly one-point-one hertz of error, and ZCD diverges all the way to four hundred hertz.
- There is no accurate winner here; high accuracy is simply not on the table.
- And worse, the rankings by RMSE, by trip-risk, and by CPU now disagree with one another. The method with the best RMSE is not the cheapest, and not the safest.

→ *Let me make the trip-risk side of that visual.*

**Key line:** Under stacked stress, there is simply no accurate winner.






# Trip-risk evidence (figure)

*Say (verbatim):*

- This is the trip-risk picture. In Scenario D, the state-space filters hold zero exposure, while the windowed methods sail past one-point-two seconds.
- But in Scenario E, look, everyone carries exposure. Nobody is safe.
- And notice the ranking here is completely different from the RMSE ranking.

→ *And you can actually watch the divergence happen, sample by sample.*

**Key line:** Trip-risk is a completely different ranking from RMSE.

# Event responses (figure)

*Say (verbatim):*

- Here it is, directly. The true frequency is the black line, and each colored line is an estimator trying to follow it.
- You can literally watch EKF and IpDFT run away on the ramp, climbing to tens of hertz, while RA-EKF just stays locked on the truth.
- This is what intermittent divergence looks like in practice.

→ *Let me zoom into one hard event to see the tracking up close.*

> **Key line:** This is the intermittent divergence, seen with your own eyes.

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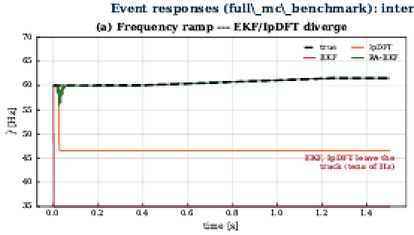
SGSMA 2026

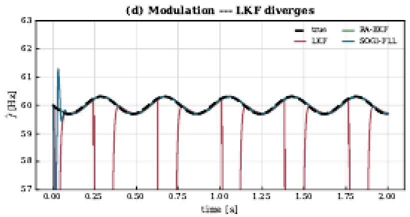
## The failures are visible in the trace, not only in the table

**RAMP DIVERGENCE**

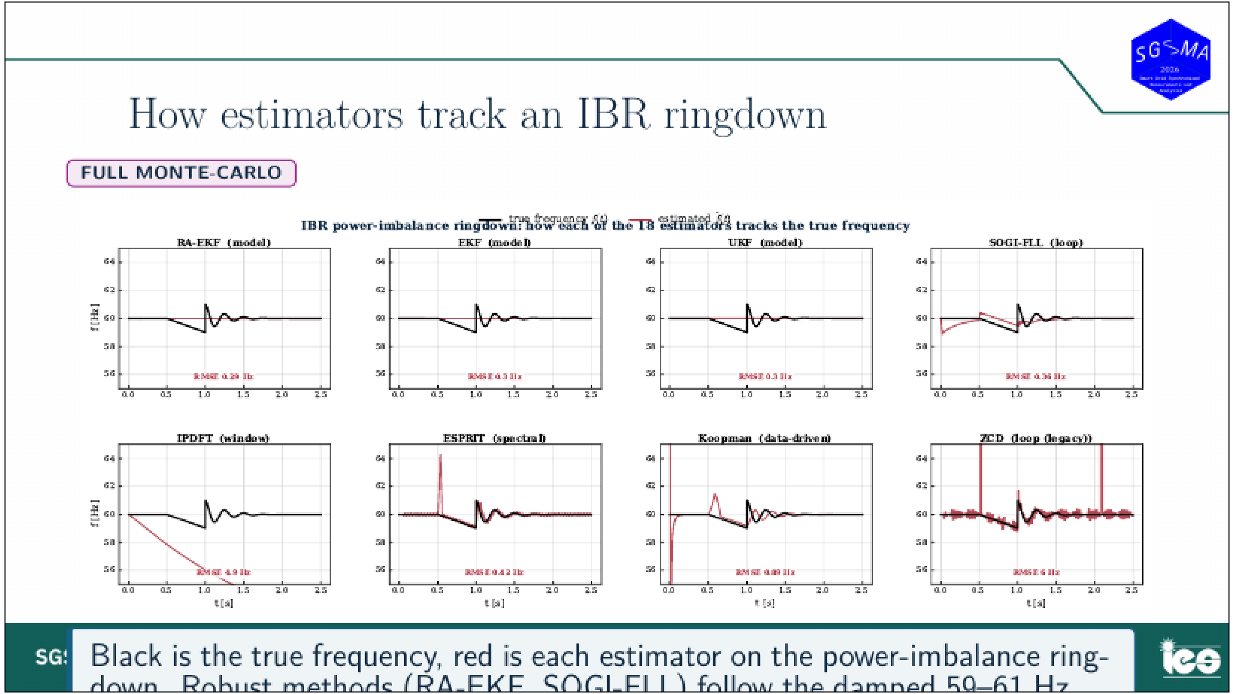



**Failure signature**
EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

**MODULATION DIVERGENCE**




**Failure signature**
LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.






# How estimators track an IBR ringdown

*Say (verbatim):*

- Here is a closer look at one hard event, the power-imbalance ringdown, across eight representative estimators.
- Black is the true frequency, the damped swing between roughly fifty-nine and sixty-one hertz; red is each estimator's own output.
- The robust methods like RA-EKF and SOGI-FLL follow the swing cleanly, while fragile ones like ZCD overshoot or simply run away.

→ *So what does all of this mean when you actually have to choose a method?*

**Key line:** Robust methods track the damped swing; fragile ones overshoot or diverge.

# Result comparison: winners depend on the question

| Question | Best-supported answer | Evidence | Boundary |
|---|---|---|---|
| Clean step/modulation accuracy | State-space + SOGI-FLL near-zero; LKF2 best on mod | Step ≈ 0; mod 0.028–0.044 Hz | Clean cases do not predict ramp/event survival |
| Ramp and RoCoF tracking | RA-EKF is the most robust ramp tracker | 0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge) | Divergence is intermittent (seed-dependent) |
| Phase jump (Scenario D) | State-space filters recover with zero trip | EKF/UKF/RA-EKF $T_{\mathrm{trip}} = 0$; IpDFT 1.21 s | Windowed/spectral carry the exposure |
| Multi-event RMSE | No accurate method — all ≈ 1.1 Hz | UKF 1.09 Hz best; ZCD 426 Hz diverges | RMSE hides trip-risk and 1000× CPU spread |
| Embedded feasibility | PLL, SOGI-FLL, EKF, RA-EKF remain plausible | 9–17 $\mu$s/sample; Koopman ∼15,000 | Python timing is a software-cost proxy |

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.





## Result comparison: winners by question

*Say (verbatim):*

- The honest summary is that the winner depends on the question you ask.
- Ask for clean steady-state accuracy, and the state-space methods and SOGI-FLL win.
- Ask for ramp robustness, and it is RA-EKF. Ask about the multi-event case, and there is no winner at all. Ask about feasibility, and the expensive methods simply drop out of contention.

→ *And that last point, feasibility, brings us straight to cost.*

**Key line:** Different question, different winner, every single time.

# Computational feasibility

| Method | Time/sample | Family | Target |
|---|---|---|---|
| SOGI-FLL | 0.0348 $\mu$s | Loop | MCU |
| SOGI-PLL | 0.0867 $\mu$s | Loop | MCU |
| EKF | 0.813 $\mu$s | Model | MCU/DSP |
| **RA-EKF** | **1.61** $\mu$s | **Model** | **DSP** |
| TFT / IpDFT | 3–4 $\mu$s | Window | DSP/FPGA |
| Koopman | 390 $\mu$s | Data-driven | CPU/GPU |
| ESPRIT | 2,838 $\mu$s | Window | CPU |

**Deployment interpretation**

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

**Key comparison**

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Source: SGSMA benchmark matrix, Sec. IV-D and Fig. 2(g)-(h)



## Computational feasibility

*Say (verbatim):*

- Cost matters, because these run on relays and embedded hardware.
- The loop and state-space methods run in nine to seventeen microseconds per sample. That is comfortably embedded-feasible.
- The spectral and data-driven methods, by contrast, are a hundred to a thousand times slower. For a real-time protection device, that is often a deal-breaker, regardless of accuracy.

→ *Put cost and risk on the same picture, and it becomes even clearer.*

**Key line:** Cost alone already rules several methods out of a relay.

# Deployment limits (figure)

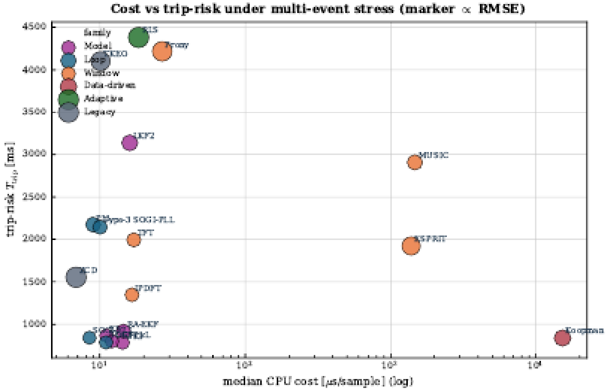
*Say (verbatim):*

- Deployment is really the combination of all of it: accuracy, trip exposure, cost, and divergence-robustness, together.
- And the encouraging thing in this picture is that the low-cost cluster, the state-space and loop methods, is also the low-trip cluster.
- The cheap methods are often also the safe ones.

→ *We can capture each method's overall balance in a single shape.*

**Key line:** The practical winners sit together, at low cost and low trip.

## Deployment limits

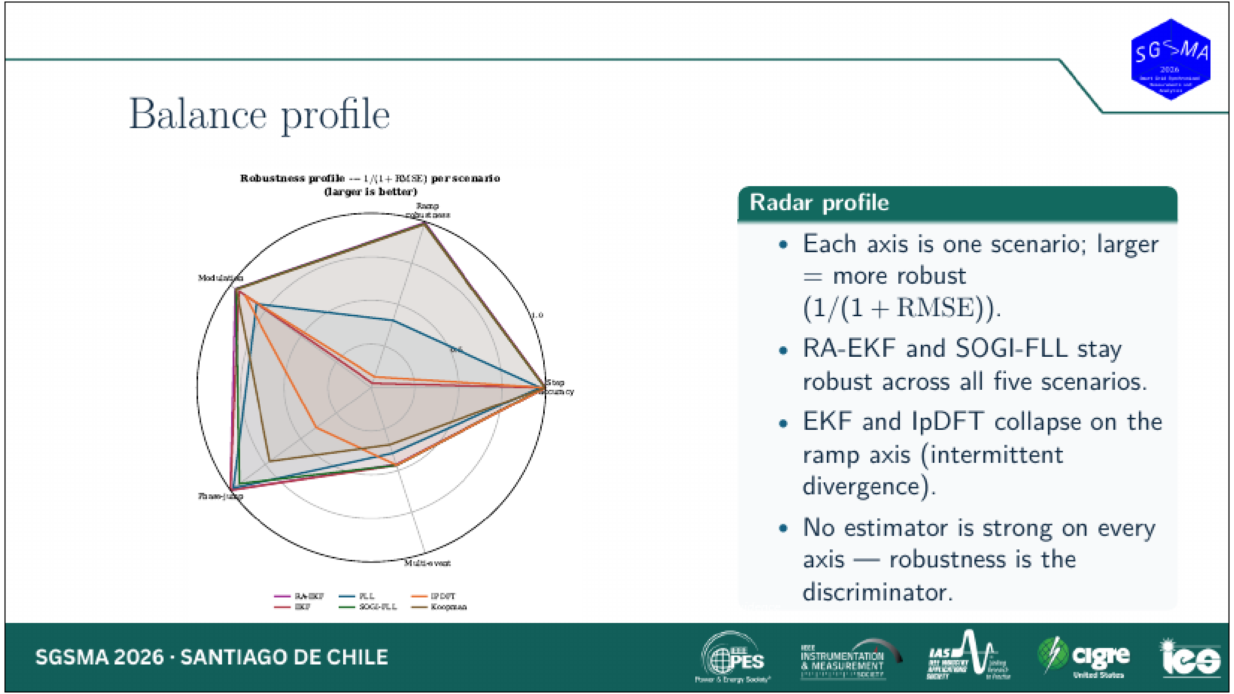



Source: OpenFreqBench expansion: artifacts/full_mc_benchmark, $n = 5$

**Deployment interpretation**

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- In the OpenFreqBench expansion, state-space + loop methods cluster at low cost ($\lesssim 17\ \mu s$) and low trip.
- Spectral/data-driven methods trade lower error for much higher CPU in this artifact.
- RA-EKF stays low-cost with no ramp divergence in this run.

# Balance profile (radar)




*Say (verbatim):*

- On this radar, each axis is one scenario, and a larger area means more robust.
- RA-EKF and SOGI-FLL stay robust on every single axis. EKF and IpDFT, in contrast, collapse on the ramp axis.
- The shape tells you the whole story at a glance.

*→ Now, this next slide is the most important one about compliance testing.*

**Key line:** No method is strong on every axis.

# Compliance heatmap: why standard tests are insufficient

| Method | Step | Ramp | Mod | Jump | Multi |
|---|---|---|---|---|---|
| Koopman | P | P | P | F | F |
| TFT | P | P | P | F* | F |
| IpDFT | P | F | P | F | F |
| SOGI-FLL | P | P | P | P | F |
| PLL | P | F | F* | P | F |
| EKF | P | F | P | P | F |
| UKF | P | P | P | P | F |
| RA-EKF | P | P | P | P | F |
| LKF2 | F* | P | P | P | F |

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/$T_{\mathrm{trip}}$/settling threshold | **F*** borderline.

**Result:** Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.





## Compliance heatmap (build-up)

*Say (verbatim):*

- This heatmap is the heart of the argument. Look at the standard columns first, step, ramp, and modulation. Almost every method passes.
- On a standard compliance test, these all look qualified.
- CLICK. But now the event columns appear, phase jump and multi-event, and watch: the same methods that passed now fail.
- Passing the standard tests told us almost nothing about event readiness.

→ *Let me distill all of this into our empirical claims.*

**Key line:** Passing the standard tests does not predict event readiness.

# Empirical claims

**Protection behavior**

**Divergence, not phase jumps, is the dominant failure mode.** On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\mathrm{trip}} = 0$.

**Metric disagreement**

**RMSE, trip-risk, and CPU select different winners.** In the OpenFreqBench multi-event run, UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

**Standard-test limitation**

**Simple-case pass/fail leaves event readiness unresolved.** The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

**Deployability**

**Feasibility changes the accuracy ranking.** In that same expansion, RA-EKF runs at ∼15 $\mu$s/sample; selected spectral and data-driven methods move far outside that cost range.

**Guardrail:** report scenario, metric, and compute envelope before naming a preferred estimator.

Source: `OpenFreqBench expansion: artifacts/full_mc_benchmark`, $n = 5$





## Empirical claims

*Say (verbatim):*

- Four empirical claims come out of the core study.
- First, divergence, not phase jumps, is the dominant failure mode.
- Second, simple pass-fail testing leaves event readiness unresolved. Third, the three metrics, accuracy, trip-risk, and cost, pick genuinely different winners. And fourth, feasibility changes the ranking again.

*→ Now the natural question is: does this pattern survive when we scale the study up?*

**Key line:** Four claims, one theme: always report the conditions.




# Expanded and ATLAS (section)

**Key line:** Does the pattern hold across families and severity? That is ATLAS.

# Expanded stress analysis

**OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario**

### Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

### ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

> Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.





## Expanded stress analysis

*Say (verbatim):*

- To test that, we scale up to a full Monte-Carlo benchmark: eighteen estimators, thirty-three scenarios, five seeds each.
- And then we add the ATLAS severity sweeps on top, which push each disturbance from mild all the way to extreme.

→ *The first question is whether the family winners stay fixed as we scale.*

**Key line:** We scale the test, we do not just repeat it.

# SGSMA benchmark: family winners

| Family | Count | RMSE leader | RMSE mean | Trip-risk leader |
|---|---|---|---|---|
| IEEE ramps | 9 | SOGI-PLL | 19.4 ± 6.4 mHz | 0 s (multiple) |
| Magnitude steps | 7 | EKF | 0.388 ± 0.143 mHz | 0 s (multiple) |
| Modulation | 3 | RA-EKF | 17.8 ± 21.4 mHz | 0 s (multiple) |
| Phase jumps | 3 | SOGI-FLL | 174 ± 67.5 mHz | ZCD: 0.0194 s |
| OOB interference | 1 | EKF | 13.6 ± 11.1 mHz | EKF / RA-EKF: 0 s |
| IBR harmonics | 3 | EKF | 70.5 ± 27.3 mHz | TFT / Prony: 0.00947 s |
| IBR ringdown | 5 | UKF | 313 ± 23.2 mHz | ESPRIT: 0.0989 s |
| IBR multi-event | 1 | PLL | 173.5 ± 66.7 mHz | IpDFT: 0.0439 s |

The paper result is conditional: SOGI-PLL leads ramps, EKF leads magnitude steps/OOB/harmonics, RA-EKF leads modulation, UKF leads ringdown, and PLL leads the composite IBR case. Trip-risk often points elsewhere.

Source: SGSMA benchmark matrix, Table II





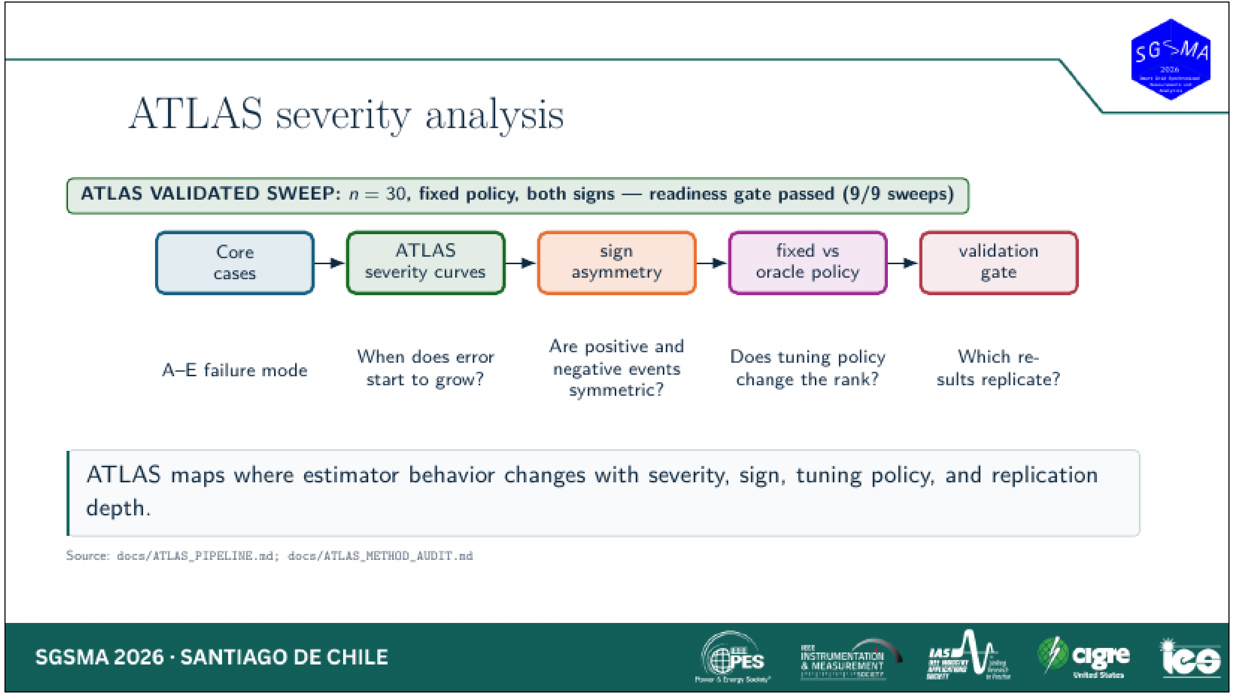
## SGSMA benchmark: family winners

*Say (verbatim):*

- And they do not stay fixed. The winner rotates from family to family.
- One method leads on the step, a different one on the ramp, another on harmonics, another on the multi-event case.
- There is no method that quietly wins everywhere when you are not looking.

→ *ATLAS lets us understand why, by sweeping the severity continuously.*

**Key line:** The leaderboard is regime-dependent, never absolute.






# ATLAS severity analysis

*Say (verbatim):*

- ATLAS sweeps each stress by severity and, importantly, by sign, under a fixed policy, at thirty Monte-Carlo runs per point.
- And I want to be clear about evidence quality here: this is now paper-grade across all nine of the required sweeps, not a quick pilot.

→ *And the headline result from those sweeps is genuinely striking.*

**Key line:** This is paper-grade severity evidence, not a pilot.

# ATLAS result: metric choice changes the answer

**ATLAS VALIDATED SWEEP:** $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

| Metric objective | Unique metric winner | Different from RMSE |
|---|---|---|
| **Trip-risk exposure** | 26 / 113 | 5 / 26 |
| **RoCoF max error** | 112 / 113 | 53 / 112 |
| **RoCoF RMS error** | 112 / 113 | 48 / 112 |

### Tie handling

Trip-risk has tied zero-exposure minima in 87/113 levels. It is an exposure screen first, not a single-winner leaderboard.

### Why this matters

For protection-adjacent use, low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

### Operational implication

RoCoF metrics change the preferred estimator in nearly half the severity map; trip-risk removes estimators with long exposure.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv





## ATLAS: metric choice changes the answer

*Say (verbatim):*

- Across a hundred and thirteen severity levels, the preferred estimator changes the moment you change the objective.
- In eighty-three of those levels, optimizing for RMSE and optimizing for trip-risk point you to two different methods.
- So the deliverable from ATLAS is not a ranking. It is a selection map: tell me your event and your metric, and it tells you the method.

→ *ATLAS also tells us something equally useful: where each method gives up.*

**Key line:** The output is a selection map, not a leaderboard.

## ATLAS result: where estimators stop being valid

**ATLAS VALIDATED SWEEP:** $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

| Stress family | Maximum tested level | Methods below 0.05 Hz | Interpretation |
|---|---|---|---|
| Magnitude step | $\pm 90\%$ sag/swell | 2 | EKF and RA-EKF survive both signs. |
| AM modulation | 10 Hz envelope | 10 | AM alone is not the hardest modulation case. |
| FM modulation | 10 Hz modulation | 0 | FM is qualitatively harder than AM. |
| RoCoF ramp | $\pm 50$ Hz/s | 0 | Severe RoCoF requires a separate qualification class. |
| Phase jump | $\pm 60^\circ$ | 0 | Single jump severity already breaks the 0.05 Hz guide. |
| Harmonics | 30% THD | 0 | Low-THD winners do not survive extreme THD. |
| Interharmonics | 20% at 75 Hz | 0 | Integer-harmonic tests are not sufficient. |
| Broadband noise | $\sigma = 0.1$ pu | 0 | Noise robustness is a first-class stress dimension. |





# ATLAS: where estimators stop being valid

*Say (verbatim):*

- For every method, ATLAS pinpoints the exact severity at which it crosses over from working to failing.
- That boundary, the point of no return, is different for every family, and knowing it is what lets you deploy a method safely.

→ *Let me now show you a few of the most surprising individual sweeps.*

**Key line:** That validity boundary is different for every family.

# AM, FM, and interharmonics split the story

**ATLAS VALIDATED SWEEP:** $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

| Stress | Winner pattern | Threshold behavior |
|---|---|---|
| **AM modulation** | EKF wins slow, reference, fast, and severe regions. | Ten methods stay below 0.05 Hz through 10 Hz envelope modulation. |
| **FM modulation** | SOGI-FLL, TFT, and Koopman split the low-rate points; ESPRIT leads from 1–10 Hz. | No method stays below 0.05 Hz at 10 Hz FM modulation. |
| **Interharmonics** | EKF leads 0.5–15%; PLL leads at 20%. | EKF crosses the 0.05 Hz guide between 5% and 8%; most families cross earlier. |

**Implication**

AM, FM, integer harmonics, and interharmonics should not be collapsed into one "distortion" label. They select different estimator families.

**Decomposition insight**

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.





## AM, FM and interharmonics

*Say (verbatim):*

- Even within modulation, the story splits in two.
- Amplitude modulation is relatively easy, and there the model-based methods win.
- But frequency modulation and interharmonics are much harder, and there the spectral methods take over. Same broad category, opposite winners.

→ *Now the handful of findings I most want you to walk away with.*

**Key line:** Even inside modulation, the winner switches.

## When the textbook filter diverges

**OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario**

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

| Estimator | seed 1 | seed 2 | seed 3 | seed 4 | seed 5 | trip-risk |
|---|---|---|---|---|---|---|
| EKF (textbook) | 0.016 | 63.7 | 0.017 | 121.5 | 0.027 | up to 1.35 s |
| RA-EKF | 0.0055 | 0.0062 | 0.0083 | 0.0045 | 0.0058 | 0 s |

### What happens

The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

### Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.





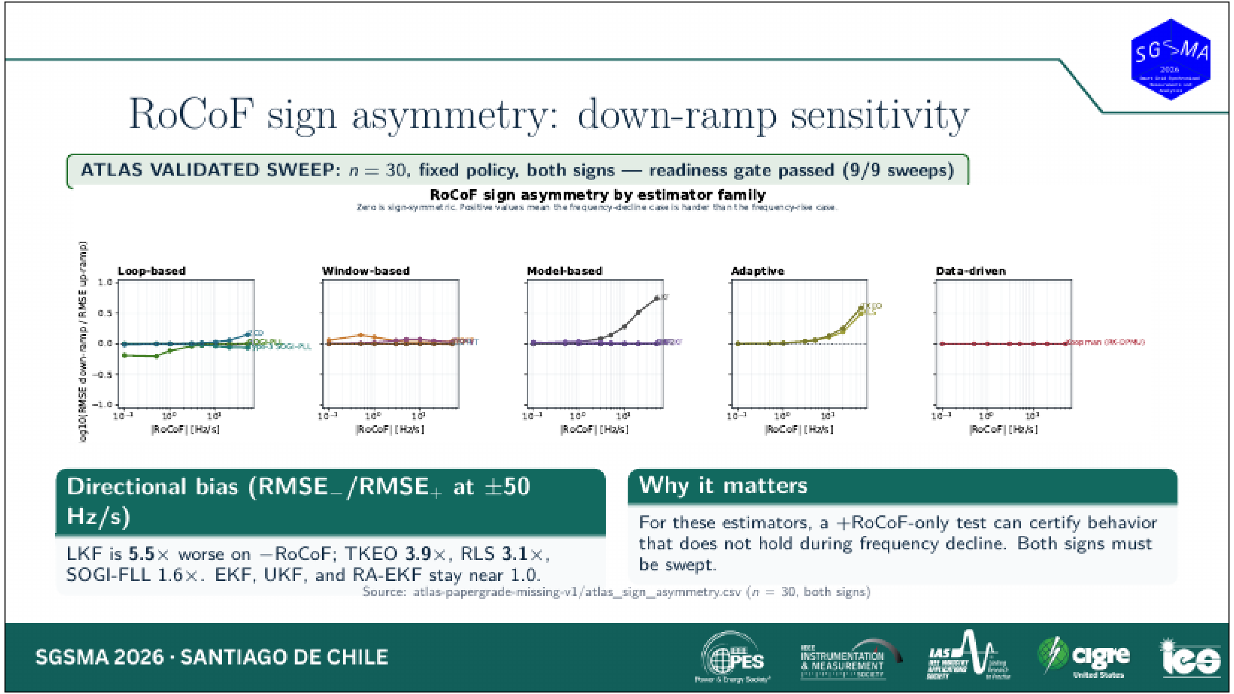
## When the textbook filter diverges

*Say (verbatim):*

- This is the divergence mechanism up close. The textbook EKF tracks most seeds perfectly well, and then, on certain seeds, it intermittently runs away to tens of hertz.
- And here is the unsettling part: it is not driven by severity. The worst seed in our set was actually the one with the gentlest ramp.
- So you cannot predict it from how hard the event looks.

→ *The next finding is about something we usually ignore: direction.*

**Key line:** The robustified RA-EKF removes the divergence on every single seed.






# RoCoF has a sign

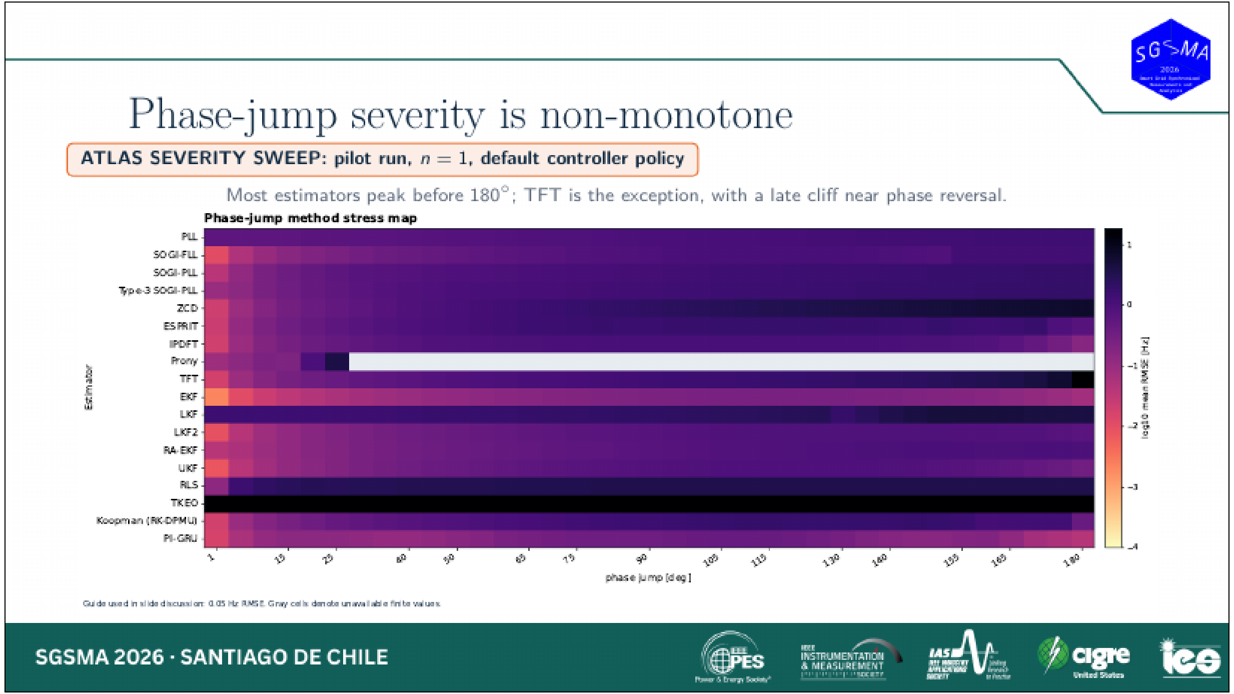
*Say (verbatim):*

- Rate-of-change of frequency has a sign, and it turns out that down-ramps are harder than up-ramps.
- The linear Kalman filter is five-and-a-half times worse when the frequency is falling. TKEO is almost four times worse.
- This matters enormously, because a falling frequency is the protection-critical direction, the one that sheds load.

→ *And severity can surprise us in the opposite direction too.*

**Key line:** A rising-only test can certify a method that fails when frequency drops.

# Phase-jump severity is non-monotone

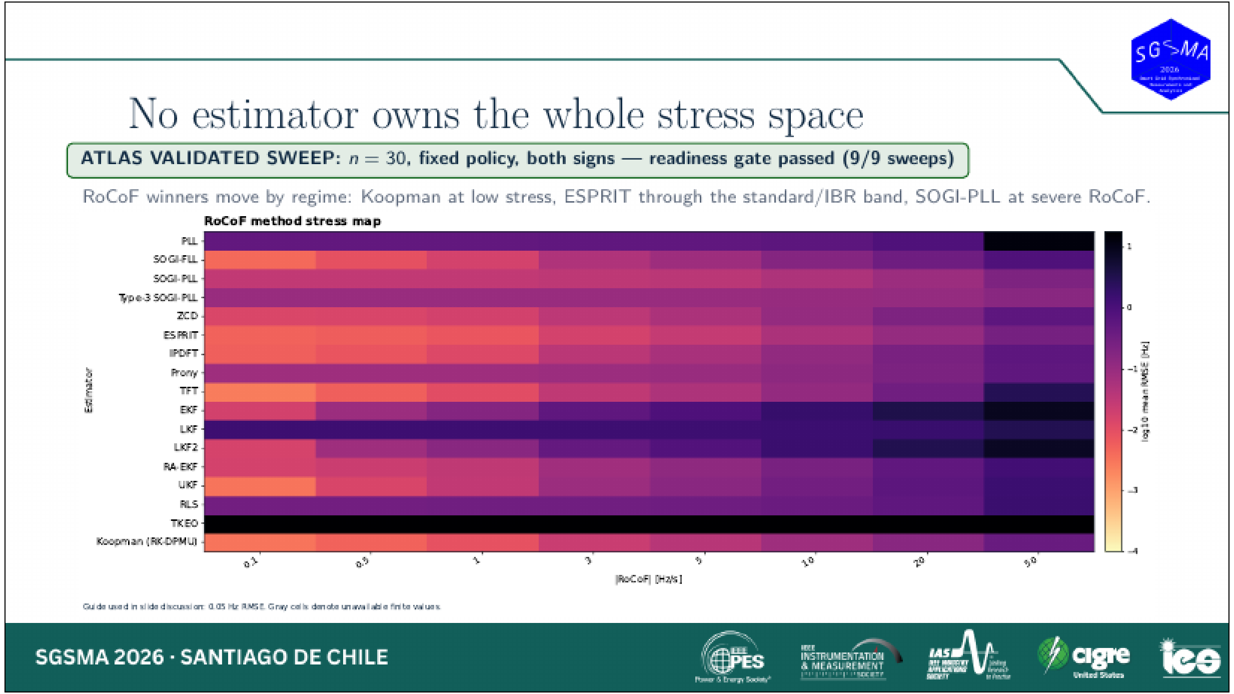



*Say (verbatim):*

- Phase-jump severity is non-monotone, which is genuinely counterintuitive. The worst case is not the largest jump.
- Most methods actually peak in error around ninety to a hundred and twenty degrees, and then recover toward a hundred and eighty, because a half-cycle jump partly cancels itself out.
- So if you test only at one nominal angle, you can completely miss the real worst case.

→ *Severity also reshuffles the winners across the whole sweep.*

**Key line:** Bigger is not always worse, so you have to sweep the range.





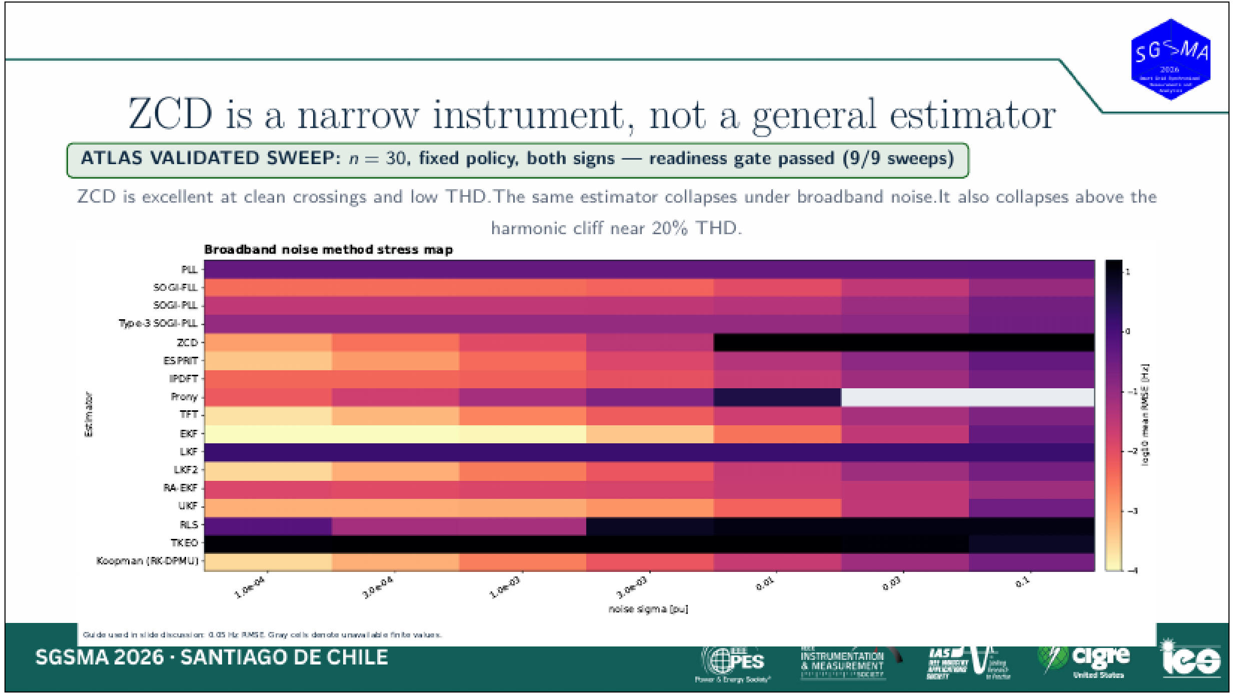

# No universal winner (map)

*Say (verbatim):*

- This map really captures the central message. The champion rotates by rate-of-change regime.
- Koopman wins at low rates. ESPRIT wins in the middle. SOGI-PLL takes over once it gets severe.
- No single column owns the whole map.

→ *And remarkably, one estimator manages to be both the best and the worst.*

**Key line:** No universal winner; the deliverable is the map itself.






## The ZCD paradox (build-up)

*Say (verbatim):*

- This is my favorite result, the ZCD paradox.
- Below fifteen percent harmonic distortion, zero-crossing detection is the single most accurate method we tested. The best of all eighteen.
- CLICK. But add broadband noise, and that very same method explodes to three thousand hertz.
- CLICK. And push past twenty percent distortion, and it collapses again, to tens of hertz.
- Best and worst, the same estimator, depending only on the stress.

→ *And the very same asymmetry shows up in voltage events.*

**Key line:** Aggregate rankings would completely hide this.

# Voltage sags and swells are not symmetric

**ATLAS VALIDATED SWEEP:** $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

| Estimator at ±90% | Sag to 0.1 pu | Swell to 1.9 pu |
|---|---|---|
| EKF RMSE | 47 mHz | 14 mHz |
| RA-EKF RMSE | 49 mHz | 30 mHz |
| ZCD RMSE | 374 Hz | 9.4 mHz |
| ZCD peak error | 2856 Hz | 32 mHz |

### Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

### What to claim

ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

### Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Source: `atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py`





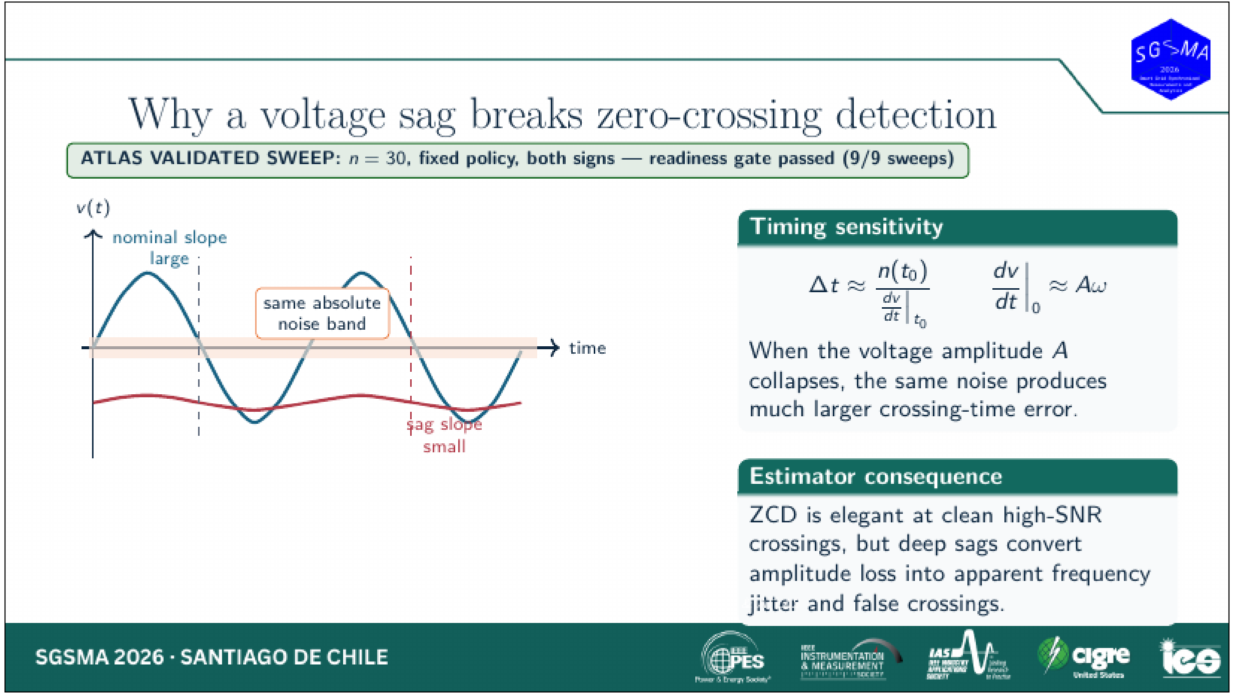
## Sags and swells are not symmetric

*Say (verbatim):*

- Voltage magnitude is not symmetric either. A deep sag is far harder to handle than an equal swell.
- Zero-crossing detection goes from millihertz of error on a swell to hundreds of hertz on a deep sag.
- The direction of the voltage event, down versus up, changes the result by orders of magnitude.

→ *And there is a clean, physical reason for that asymmetry.*

**Key line:** Direction matters, so you have to sweep both signs.






# Why a sag breaks zero-crossing

*Say (verbatim):*

- The mechanism is simple physics. The timing error of a zero-crossing scales with the noise divided by the amplitude times the frequency.
- A deep sag shrinks the amplitude, which flattens the slope of the waveform right at the crossing.
- And once the slope is flat, even small noise creates false crossings. It is not a coding bug; it is the geometry of the signal.

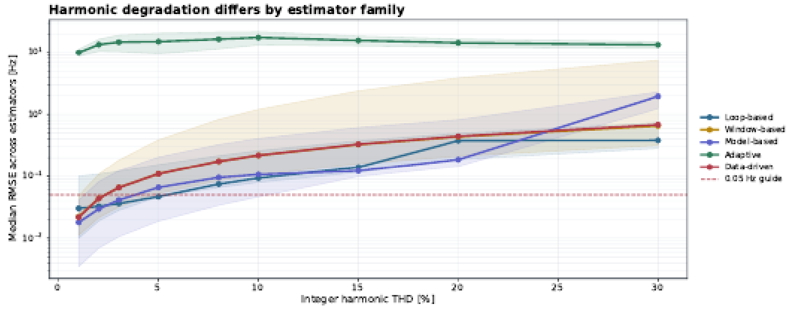
→ *Harmonics tell a similar, family-by-family story.*

**Key line:** This is physics, not a software bug.

# Harmonic degradation differs by estimator family

**ATLAS VALIDATED SWEEP:** $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Family medians cross as THD increases; low-THD leadership does not predict high-THD survival.




Band: interquartile range across estimators in the same family. Source: OpenFreqBench ATLAS harmonics sweep.



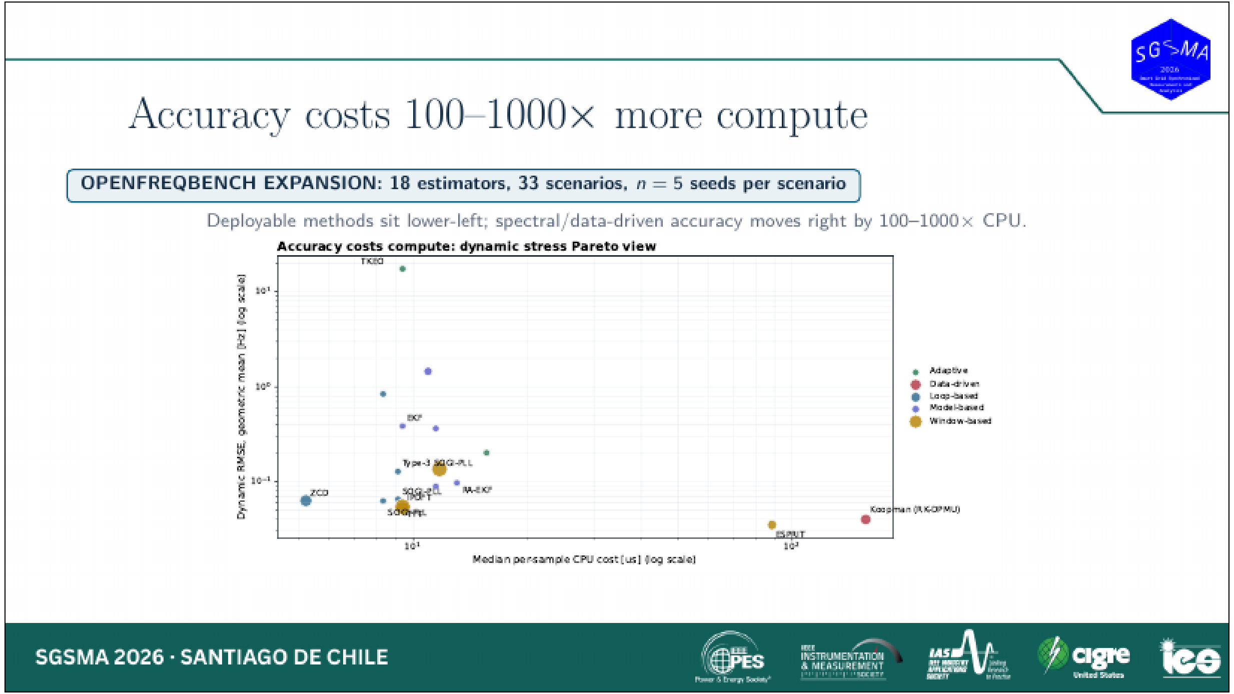
## Harmonic degradation by family

*Say (verbatim):*

- Harmonic degradation also depends strongly on the family.
- Loop methods like ZCD win at low distortion, but they collapse once you go above roughly fifteen percent.
- Window methods degrade much more gently, and they actually end up leading at extreme distortion. So low-distortion accuracy does not predict high-distortion survival.

→ *And cost is the final dimension to bring in.*

**Key line:** Low-distortion accuracy does not predict high-distortion survival.






# CPU-accuracy Pareto

*Say (verbatim):*

- On the cost-accuracy Pareto front, the spectral and data-driven methods do buy you a little extra accuracy.
- But they pay for it with a hundred to a thousand times more computation.
- For relay-class, real-time timing, that trade is almost never worth it.

→ *So, with all of that on the table, what do we actually deploy?*

**Key line:** For relay-class timing, that trade is usually not worth it.




# Findings and deployment (section)

**Key line:** So what do we actually deploy?

## Three findings

**1. Metric-conditioned**
The SGSMA matrix and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

**2. Stress-bounded**
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

**3. Robustness over point accuracy**
The OpenFreqBench ramp seeds expose silent divergence; RA-EKF removes that failure mode in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.





# Three findings

*Say (verbatim):*

- Three findings sum up the work.
- First, there is a clear latency-versus-stress trade-off; the most accurate methods are often the slowest.
- Second, robustness beats point accuracy. RA-EKF, with its explicit rate-of-change state, stays stable exactly where the textbook EKF and IpDFT diverge.
- And third, compliance tests, on their own, are simply not enough for inverter-dominated events.

→ *Let me be honest about the limits of what we have shown.*

**Key line:** Robustness, not accuracy, is the deployment discriminator.

# Validity limits and next steps

**Validation limits**

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

**Validation program**

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

> These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.





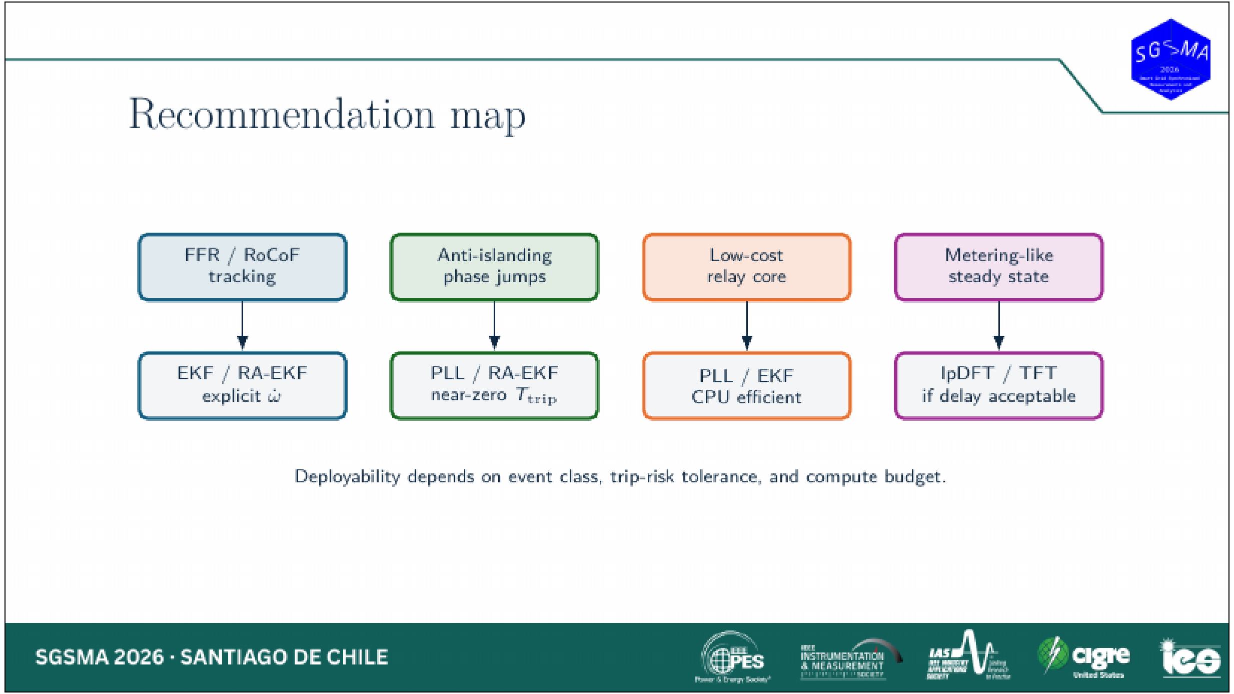
## Validity limits and next steps

*Say (verbatim):*

- On validity, I want to be transparent. The core scenario means are at five seeds, and the ATLAS severity sweeps are paper-grade at thirty.
- The next step is journal-grade, at a hundred runs, and adding the subspace and machine-learning methods to the full sweep.
- We are deliberately explicit about what is diagnostic and what is paper-grade evidence.

→ *From there, the work becomes a concrete recommendation.*

**Key line:** We are explicit about what is diagnostic and what is paper-grade.





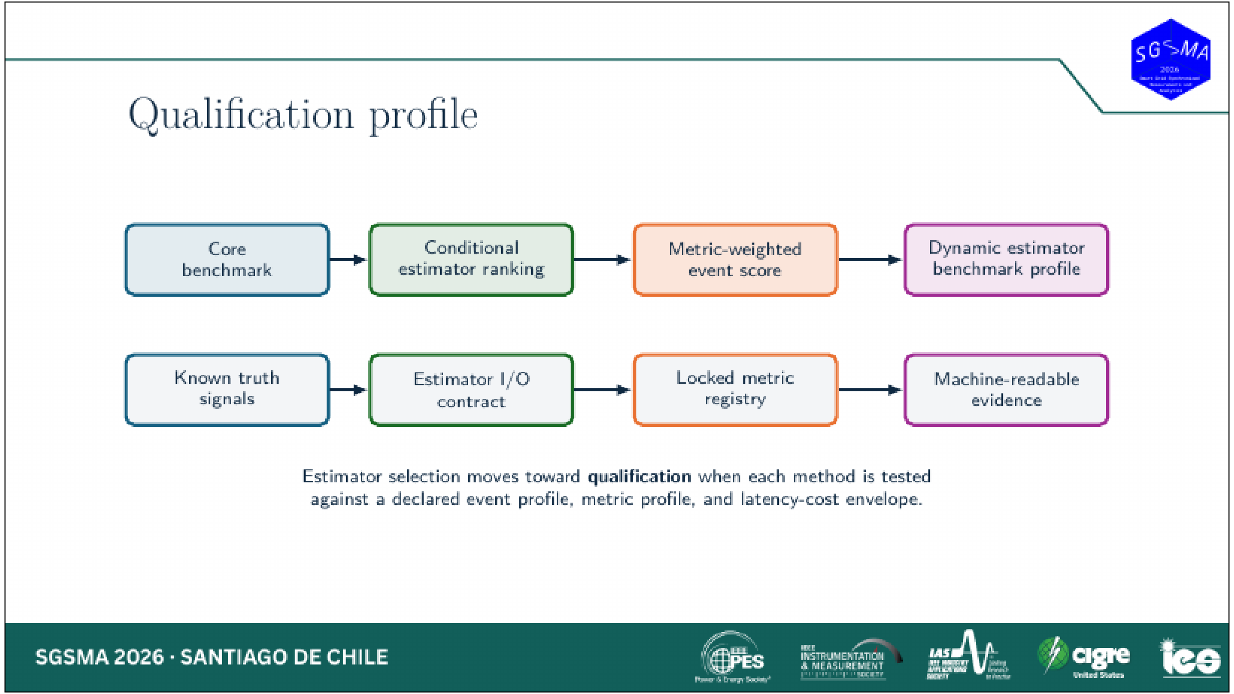

## Recommendation map

*Say (verbatim):*

- So here is our recommendation, and notice it is a map, not a single answer.
- For fast frequency response and anti-islanding, RA-EKF is the strongest, best-supported choice in our benchmark.
- PLL and EKF make solid low-cost baselines. And IpDFT or TFT are good where some observation delay is acceptable.

→ *We then wrap that into a qualification idea.*

**Key line:** We deliver a recommendation map, not a single ranking.






# Qualification profile

*Say (verbatim):*

- A real qualification profile, in our view, should test transient recovery, trip exposure, compute feasibility, and standard-case RMSE, all together, not one at a time.
- Because, as we have seen, a method can pass any one of these and fail the others.

→ *And it should be reported in a standard, comparable form.*

**Key line:** Qualify on all dimensions at once, not one at a time.

# Standardization profile

## Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

## Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

## Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

## Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.





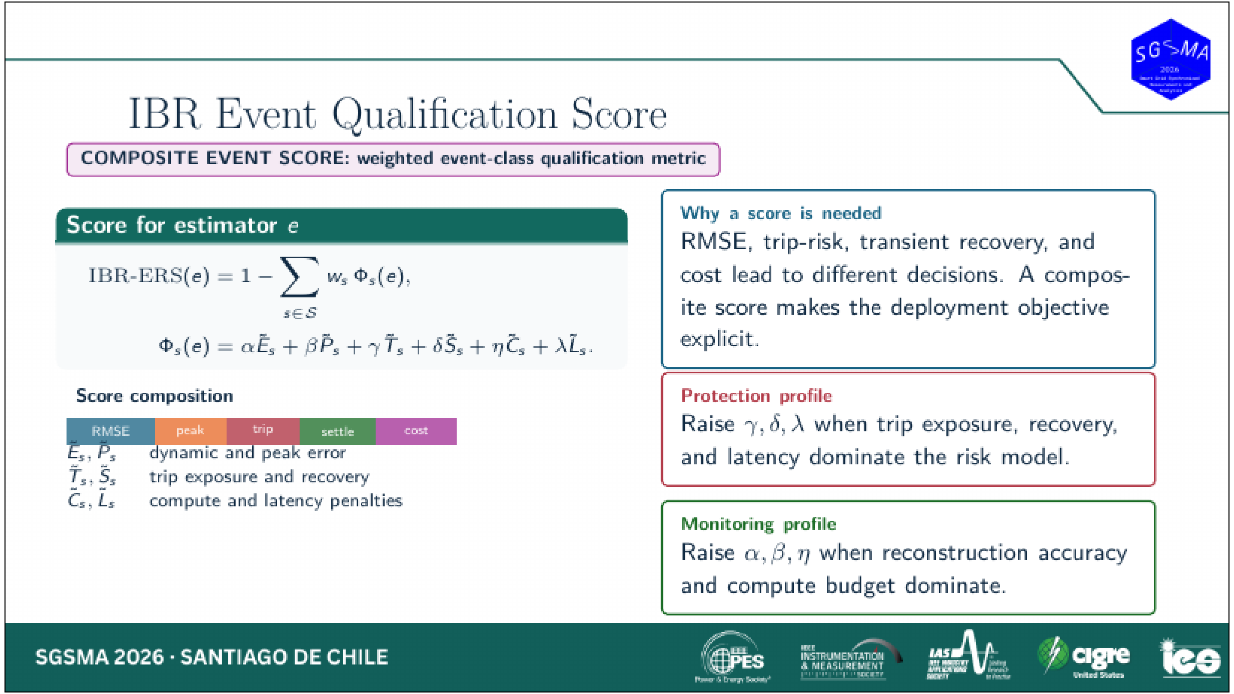
## Standardization profile

*Say (verbatim):*

- So we propose reporting these dimensions explicitly and consistently, so that any two estimators can be compared on truly equal terms.
- That is what turns a one-off study into something the community can build on.

→ *And when you genuinely need just one number, here is how we do it.*

**Key line:** Make the comparison reproducible and fair.





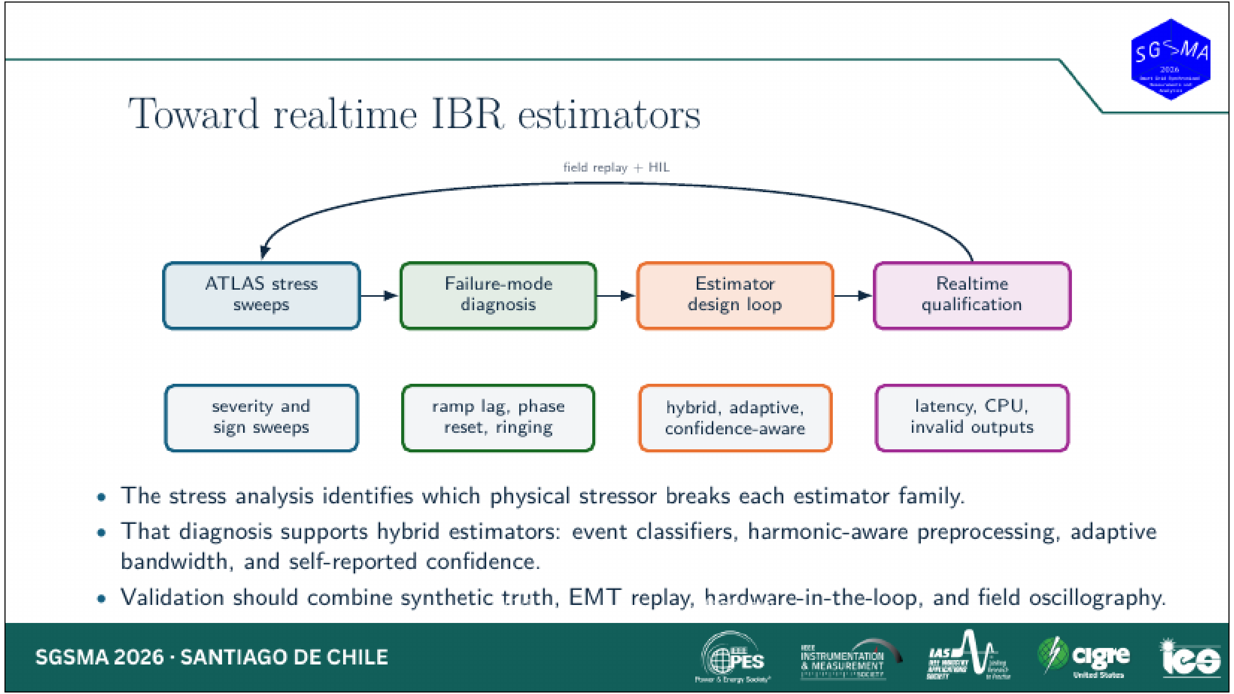

# IBR Event Qualification Score

*Say (verbatim):*

- When a single number is unavoidable, for instance in a procurement spec, we define a composite, weighted, event-class qualification score.
- It rolls the dimensions into one figure, but it does so transparently, so you can always see what went into it.

→ *Finally, let me point to where this work is heading.*

**Key line:** One number when you need it, without hiding the dimensions.






## Toward realtime IBR estimators

*Say (verbatim):*

- Looking ahead, there is a clear roadmap.
- Real-time estimators on hardware. Integrating recorded Simulink microgrid fault data. Adding modern and machine-learning methods. C++ acceleration for the heavy methods. And co-simulation with ANDES for closed-loop studies.

→ *And with that, let me close.*

**Key line:** This is where the benchmark goes next.

# Validation agenda and conclusion

## Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

## Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

> **Result:** estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.





# Validation agenda and conclusion

*Say (verbatim):*

- So, to conclude. The single most important message is that there is no universal frequency estimator.
- The defensible choice always depends on the event you expect, the metric you care about, and the compute budget you have.
- And what OpenFreqBench provides is a reproducible, open way for the community to make that choice on evidence, rather than on habit.

> **Key line:** Thank you very much for your attention. I would be very glad to take your questions.